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The impact of artificial intelligence (AI) on employment and economic growth in BRICS: Does the moderating role of governance Matter?

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ABSTRACT

The BRICS countries, namely Brazil, Russia, India, China, and South Africa, are committed to achieving and sustaining Sustainable Development Goal (SDG) 8 of the United Nations, which focuses on sustainable economic growth and productive employment and decent work for all. However, the empirical research question on how to effectively harness the use of artificial intelligence (AI) in the presence of governance dynamics is yet to be addressed in the literature, particularly concerning the BRICS economy. Hence, it is crucial to examine the impact of AI investment on employment and economic growth in the BRICS countries from 2012 to 2022, taking into account the moderating role of overall governance as well as specific governance indicators such as political governance, institutional governance, economic governance etc. Cross-Sectional Augmented Autoregressive Distributed Lag (CS-ARDL) technique is used to analyze the data. The findings of the study suggest a long-run equilibrium relationship among the variables analyzed in both the employment and growth models. The causality results for our main variables of interest differ in the employment-growth models. Based on the CS-ARDL results, the study recommends that BRICS governments and policymakers prioritize and enhance the integration of AI into their governance systems to promote employment and stimulate growth in both the short- and long-term. However, the study cautions against overlooking interaction between AI and general governance as it did not demonstrate support for growth, it is crucial to implement robust measures to mitigate any potential negative effects arising from AI's interaction with political and institutional governance. Therefore, the study proposes the formulation of AI-friendly governance policies among BRICS nations to stimulate employment and growth, recognizing the embryonic status of AI as a pivotal technology in the Fourth Industrial Revolution.

1. Introduction

The BRICS countries, namely Brazil, Russia, India, China, and South Africa, represent a diverse group of developed economies characterized by rapid technological advancements, varying levels of employment, growth and governance development. And they often work together to address economic challenges, promote free trade, and collaborate on global issues like climate change, unemployment, security, and development (Lowe, 2016). As these nations strive to promote employment, economic growth and development, the interplay between artificial intelligence (AI) and governance,[1] and their impact on employment and growth becomes a topic of significant interest and importance. Understanding the dynamics between these variables in this context is essential for formulating effective policies and strategies to harness the benefits of AI-driven advancements in BRICS economies.

This study's position of inquiry is driven by four primary considerations, namely: (i) The emerging field of AI has the potential to revolutionize various aspects of society, including governance/institutions, employment, growth, development etc.; (ii) BRICS' challenge of sustaining United Nations Sustainable Development Goal (SDG) 8, which

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[1] Based on the literature and to the best of our knowledge, we utilized Kaufmann et al., (2010,2011) six governance indicators (see Table 1) for this study. These indicators are widely recognized and can be further categorized into: (i) "institutional governance, (ii) "political governance", and (iii) "economic governance". Asongu and Nwachukwu (2016) define each dimension in accordance with Kaufmann et al., (2010,2011) study as follows: "*(i) "institutional governance" which is the respect of the State and citizens of institutions that govern interactions between them (measured with corruption-control and the rule of law); (ii) "political governance" which is the election and replacement of political leaders (proxied with political stability/no violence and voice and accountability) and (iii) "economic governance", which is defined as the formulation and implementation of policies that deliver public commodities (measured with government effectiveness and regulation quality)" (p. 135).*

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focuses on sustainable economic growth and productive employment and decent work for all; (iii) the vital role of governance quality in either promoting or impeding employment and economic growth; and (iv) the existing research gaps in the empirical literature of AI-governance-employment-growth nexus that demand thorough examination.

AI, with its rapid advancements in recent years, has the potential to transform industries, societies, and economies. AI encompasses a range of technologies and applications that simulate human intelligence and perform tasks autonomously. Its potential benefits in areas such as automation, data analysis, and decision-making make it a powerful tool for enhancing productivity, efficiency, innovation and human well-being (Makridakis, 2017; Saba and Pretorius, 2024). AI is revolutionizing human interactions and business practices, potentially igniting a fourth industrial revolution (Lu, 2021). Traditional production factors

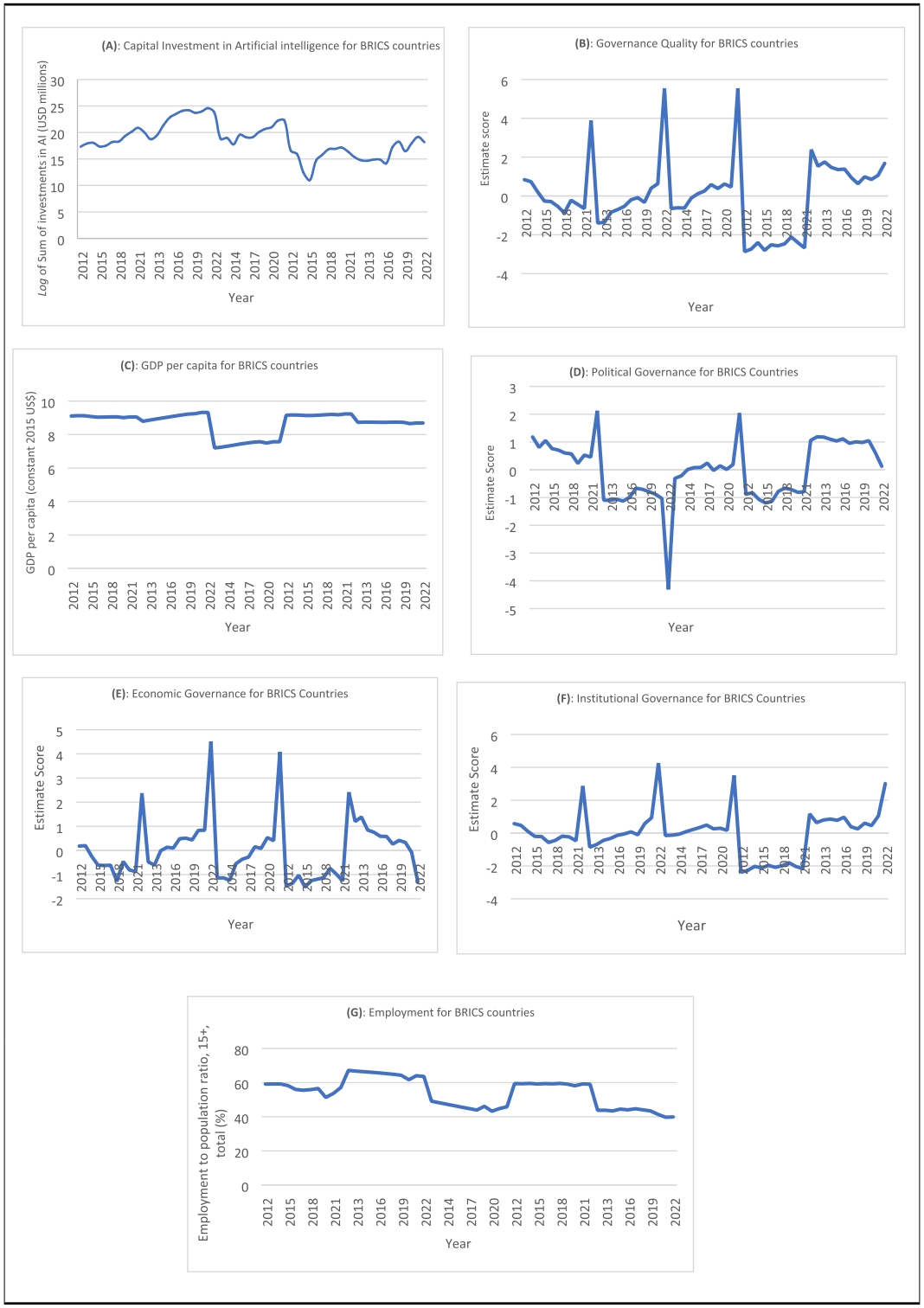

Fig. 1. (A): Capital Investment in Artificial intelligence for BRICS countries; (B): Governance Quality for BRICS countries; (C): GDP per capita for BRICS countries; (D): Political Governance for BRICS Countries; (E) Economic Governance for BRICS Countries; (F) Institutional Governance for BRICS Countries; (G) Employment for BRICS Countries.

like physical capital and labor may no longer be the primary drivers of growth (Acemoglu and Restrepo, 2018a; Lu, 2021). The prevailing belief is that AI will play a pivotal role in future employment and growth, and in contrast to conventional machines that replaced human and animal labor for specific tasks, AI inputs have the capacity to fundamentally transform the nature of work (Acemoglu and Restrepo, 2018b; Lu, 2021). AI is widely regarded as crucial for future growth (Lu, 2021). For example, research on the US economy by Fernald and Jones (2014) found that R&D plays a significant role in economic growth. They suggested that AI's potential to replace workers to some extent might contribute to future growth.

Governance quality, encompassing elements such as rule of law, transparency, political stability, regulatory frameworks etc., sets the foundation for economic activities, investment, entrepreneurship and economic development within a country (Chhabra et al., 2023). They provide stability, predictability, and trust, attracting both domestic and foreign investments (Chhabra et al., 2023). *On the one hand*, studies have shown a strong correlation between governance quality and inclusive growth, particularly when it comes to raising living standards and stabilizing the foundation for social change through effective resource allocation in a country (Fosu, 2013; Anyanwu and Erhijakpor, 2014; Efobi, 2015). *On the other hand*, authors have hypothesized that stronger growth in the future could result from the potential for AI to partially replace workers with robots. Aghion et al. (2018), Acemoglu and Restrepo (2018b), Cai and Chen (2019), Lu (2021), Braganza et al. (2021), Damioli et al. (2021), Hémous and Olsen (2022), Yang (2022), Guliyev (2023), among others, represent recent studies exploring the potential effects of AI on either employment or the growth process. Although the contributions of AI and institutional/governance quality to welfare, growth, and development are well-documented in the empirical literature (see for example, Fosu, 2015; Asongu and Nwachukwu, 2016; Acemoglu and Restrepo, 2018b; Lu, 2021), the role of governance quality/dimensions/indicators in mediating these effects in BRICS nations remains unexplored in the expanding literature on AI-related employment and growth. Hence, the rationale for this study at the BRICS economic-bloc level.

Furthermore, Fig. 1 below presents a graphical representation of the time paths for our main variables of interest in this study, focusing on the BRICS countries. From the preliminary information conveyed by the visual graph, it is apparent that the five variables did not demonstrate a similar time path throughout the study period. This observation further underscores the importance of investigating the interrelationships between the variables depicted in Fig. 1. To the best of the authors' knowledge, there is a dearth of studies investigating the interplay relationships between these variables in the context of BRICS countries.

Therefore, the primary objective of this study is to address this research gap in the literature. Therefore, the research question that this study seeks to address are: (i) What are the causal relationships among the main variables of interest in the BRICS economy?; (ii) what are short- and long-run impact of artificial intelligence (AI) and governance quality/dimensions/indicators on employment and economic growth in BRICS economy?; (iii) what are short- and long-run mediating effects of governance quality/dimensions/indicators on the impact of artificial intelligence on employment and economic growth in BRICS economy?; and (iv) what policy implications can be derived from the findings regarding the relationships between the key variables of interest?

These are the research questions that inspire the study's objectives and contributions. Therefore, the broad objective for this study seeks to explore how AI complements governance quality/dimensions/indicators in order to promote employment and economic growth in a panel of BRICS economies over the period 2012–2022. While specifically, (i) the study investigates the causal relationships among artificial intelligence (AI), governance quality/dimensions/indicators, employment and economic growth; (ii) to investigates the short- and long-run impact of artificial intelligence (AI), governance quality/dimensions/indicators on employment and economic growth; (iii) to investigates the short- and long-run impact of the interaction between AI and governance quality/dimensions/indicators employment on economic growth; (iv) we used principal component analysis approach to compute the governance quality/dimensions variable from six governance indicators; and (vi) to discuss policy implications from the results of the study. The empirical results can inform evidence-based policy decisions regarding the adoption and implementation of AI technologies in governance frameworks to support growth in BRICS economy.

Our research on BRICS contributes to the existing literature in several ways compared to previous empirical studies. Firstly, it investigates the causal relationships among artificial intelligence (AI), employment, governance quality/dimensions/indicators and economic growth. Secondly, it investigates the short- and long-run impact of artificial intelligence (AI), governance quality/dimensions/indicators on employment and economic growth. Thirdly, investigates the short- and long-run impact of the interaction between AI and explanatory variables such as governance quality/dimensions/indicators on employment and economic growth. Lastly, it discusses policy implications from the results obtained. We also consider the issues of homogeneity and cross-sectional dependence (CD) in the series under the unit root and cointegration techniques. Most panel series in the literature assume that the residual terms are unrelated, which ignores the issues of CD between countries. Accounting for the issues of CD in the series is essential as BRCIS countries possess varying levels of economic tiles. We applied the Dumitrescu-Hurlin (2012) panel causality technique that considers heterogeneity among countries. Additionally, to the best of our knowledge, this is the first study to simultaneously employ the novel Cross-Sectional Autoregressive-Distributed Lag (hereafter, CS-ARDL) technique proposed by Chudik and Pesaran (2015). We primarily relied on the CS-ARDL approach, given its advantages over the others. The CS-ARDL method has several advantages over other panel data estimators. It yields robust results in the presence of cross-sectional dependence and can be applied to series with different orders of integration, including *I(0)*, *I(1)*, or both. Additionally, it produces accurate results in cases of endogeneity, weak exogeneity and allows for both pooled, mean group, and pooled-mean group estimates based on the homogeneity or heterogeneity of slope coefficients.

This study bears academic and policy importance in four dimensions. Firstly, it emphasizes the joint impact of AI and explanatory variables (e.g., governance quality/dimensions/indicators) on employment and growth within the BRICS economy. Secondly, it delves into the interconnectedness among AI, governance quality, employment, growth, and governance dimensions/indicators, shedding light on how BRICS nations could formulate their AI and governance policies and structures. The primary theoretical contribution of this study is that it briefly shows that modified Cobb-Douglas production function (Cobb and Douglas, 1928) following previous empirical studies could be used to investigate this study's objectives.

The organization of this paper is as follows: Section 2 presents the literature review. Section 3 presents the methodology and data. Results from the empirical analysis are presented and discussed in Section 4, while Section 5 concludes the study alongside policy recommendations.

2. Literature review

The relationship among AI, governance quality, and economic growth is a complex and evolving area of study. This literature review aims to provide an overview of the existing research exploring the connections between these variables. Consequently, this section is categorized into two parts: (i) empirical reviews addressing the AI-growth nexus, (ii) AI-employment nexus and (ii) examinations of the governance/institutional quality/indicators-growth nexus.

AI-Growth Nexus

AI, represented by industrial robots or investment in AI, has emerged

as a transformative resource and technology, propelled by groundbreaking advancements (Zhao et al., 2022). With the advent of deep learning, AI has swiftly found applications across various economic and societal domains, expanding its reach and impact. This pervasive integration of AI has enhanced its capacity to permeate society, leverage resources, and foster continuous growth (Zhao et al., 2022). Significantly, AI has emerged as a pivotal catalyst for technological progress, presenting novel opportunities for growth and societal advancement. The roots of AI can be traced back to the Dartmouth Conference held in 1956, during which it was formally introduced as the discipline focused on the development of intelligent machines and computer programs (Zhao et al., 2022). This landmark event marked the inception of extensive research efforts in the field, setting the stage for subsequent advancements (Moor, 2006).

The existing body of literature on AI predominantly centers around topics such as economic growth and development, employment, total factor productivity (TFP), technical innovation, income distribution, industrial integration model (Goos et al., 2009; Hémous and Olsen, 2022; Zhao et al., 2022; Aghion et al., 2018; Arntz et al., 2016; Kromann et al., 2020; Brynjolfsson et al., 2019; Iain et al., 2019; Purdy et al., 2017). Recently, a growing number of studies have begun to investigate the impact of AI on growth. Notably, Graetz and Michaels (2018) identified TFP as a significant transmission channel through which AI affects growth. Additionally, Acemoglu and Restrepo (2018a, b) posited that AI has the potential to effectively address the challenges posed by aging populations and enhance economic growth by augmenting TFP. In their study, Aghion et al. (2018) utilized Baumol's "cost disease" and Zeira's models to investigate the potential impact of AI on the growth process. They integrated AI into the production function and aimed to stabilize both the capital share and per capita GDP growth over time. The study's findings indicate that: (i) Baumol's cost disease leads to a reduction in the GDP share of the sectors under examination (namely, manufacturing and agriculture), but this reduction is counteracted by the overall automation rate of the economy; and (ii) Depending on its introduction, AI can either temporarily or permanently accelerate growth. Qiu et al. (2021) presented a novel growth model that integrated technological progress and GDP. The findings of the study indicate that countries along the "Belt and Road" initiative have the potential to experience substantial improvements in their GDP through technological advancements. The recent study of Lu (2021) developed the three-sector endogenous growth model to examines the growth and welfare effect of AI. The findings show that the advancement of AI has the potential to enhance growth during the transitional dynamics phase and can contribute to short-term utility gains for households when AI accumulation results from increased productivity in the goods goods sector or AI sector. However, if AI accumulation is primarily driven by firms substituting human labor with AI, it may have adverse effects on short-term household utility. Bandari (2019) examines the revenue growth effects of AI for 391 small businesses in developing countries. The study's findings underscored the significance of AI in small businesses and indicated that leveraging AI applications can yield favorable outcomes in terms of revenue growth. Hémous and Olsen (2022) developed an endogenous growth model incorporating automation and horizontal innovation. Their findings provide a novel perspective on the recent income distribution trends, demonstrating that these trends can be internally explained within their model.

Governance/institutional quality/indicators-Growth Nexus

Over the years, extensive research has examined various factors that contribute to short- and long-term economic growth and development at different levels, including country-specific, regional, economic blocs, and global levels. These studies have identified factors such as trade openness, technology, government size, and income distribution as significant determinants of growth (see, for example, the pioneering work by Barro, 1991, and the study conducted by Levine and Renelt, 1992, among others). Furthermore, emerging growth theories have put forth the idea that quality of governance/institutions may also play a critical role in understanding the variations in economic performance across countries and regions (Butkiewicz and Yanikkaya, 2006; Nawaz et al., 2014; Salman et al., 2019). For example, the studies by Dollar and Kraay (2000) and Butkiewicz and Yanikkaya (2006) support the fact that the rule of law promotes growth. Hashim et al. (2011) examines economic performance effect of governance in 27 SSA countries for the period between 1984 and 2003. The findings of the study reveal that governance variables play a crucial role in the process of economic performance, while the control variables used in the study exhibit a minimal impact. Fayissa and Nsiah (2013) investigate the governance-economic growth nexus for African countries and find that good governance is crucial to differences in the growth of African countries, which further depends on the level of income of the countries.

Huang and Ho (2017) examines the governance-growth nexus across 12 Asian countries from 1996 to 2014, employing the frequency domain causality technique. Their results indicate that various governance indicators have a more pronounced impact on economic growth in 'Not Free' countries compared to 'Free' and 'Partly Free' nations. Sabir and Qamar (2019) examines the governance quality-inclusive growth nexus for developing Asian countries spanning 1996 to 2017 by using the SGMM approach. The empirical results reveal inclusive growth positive effects of fiscal policy and governance. Singh (2022) examines the governance-growth nexus for BRICS countries over the period 1997–2015. The study applied the FMOLS and Dumitrescu and Hurlin panel causality approaches. The results indicate that over the long term, governance has a positive impact on growth. However, in the short term, a certain level of economic development is necessary for institutions to function effectively. Pradhan, et al. (2023), examine the governance/institutional quality-growth nexus with other variables for lower-income countries (79) between 2005 and 2022. They utilize the vector error correction model (VECM) approach and finds that governance quality promotes growth.

AI-Employment Nexus

Research on the impact of AI on employment structures suggests significant transformations in the labor market and these studies includes Smith and Anderson (2014), Acemoglu and Restrepo (2018), Susskind and Susskind (2015), Arntz et al. (2016), Grartz and Michaels (2018), Humlum (2019), among others. For example, Smith and Anderson (2014) found that 48 % of tech experts believed robots could replace most standardized labor, risking unemployment for many except highly skilled workers. Acemoglu and Restrepo (2018) argued AI would create new, complex jobs but eliminate traditional labor-intensive ones. Susskind and Susskind (2016) anticipated heightened technical unemployment and the replacement of routine labor by machines. Arntz et al. (2016) estimated a 9 % job replacement rate by robots in 21 countries. Grartz and Michaels (2018) observed robots disproportionately affecting low-skilled tasks.

Humlum (2019) noted increased unemployment among traditional production workers and rising demand for skilled workers due to industrial robots. Cai and Chen (2019) predicted a 'skilled technological revolution' in the fourth industrial revolution, deepening structural contradictions in labor supply and demand. Using 232 survey approach, Braganza et al. (2021) study investigates the effects of AI adoption on workers' psychological contract, engagement, and trust, in the context of the United Nations' SDG 8. The study finds that: (i) Psychological contracts have a significant positive effect on job engagement and trust. However, with the adoption of AI, this positive effect significantly decreases; (ii) the study introduces the concept of an "Alienational" psychological contract, suggesting that AI adoption fosters a new type of contract that detracts from the nature of decent work; and (iii) AI adoption negatively impacts employee engagement due to work uncertainty but does not significantly impact employees' trust in the

company.

Damioli et al. (2021) investigated the nexus between AI and labor productivity, analyzing a global sample of 5,257 companies with a history of at least one AI patent filing. The findings reveal that AI patent applications significantly enhance labor productivity, with a more pronounced impact observed in Small and Medium Enterprises (SMEs) and the services industries. Yang (2022) explores the effects of AI technology on firm productivity and employment. The study encompasses firm performance in Taiwan's electronics industry and spans from 2002 to 2018 and uses a firm-level dataset alongside a keyword-matching method for AI patent identification. The key findings indicate that AI technology is positively associated with both productivity and employment, while also significantly altering firms' workforce compositions, particularly reducing the share of labor with college-level education and below. Guliyev (2023) investigates the impact of AI on unemployment in 24 high-tech countries from 2005 to 2021, analyzing the correlation between AI interest (via Google Trend Index) and unemployment rates. Using dynamic panel data and GMM-system estimation approach, the study finds AI reduces unemployment and confirms its '*displacement effect*'.

Research Gap

Despite empirical literature on AI-growth nexus and governance/institutional-growth nexus, to the best of our knowledge, no studies have examined how governance quality/dimensions/indicators can distinctively complement AI-employment-economic growth nexus in BRICS economy using the novel CS-ARDL approach. It is on this basis that we embark on this study for the purpose of recommending policies that will help BRICS countries achieve the SDG 8, which focuses on sustainable economic growth and productive employment and decent work for all as mentioned earlier.

3. Methodology

3.1. Empirical strategy

The preliminary empirical strategy used in this study include principal components approach/analysis (PCA), descriptive analysis, scatter plot (graph), panel unit root test (first-and second-generation), slope homogeneity test, cross-sectional dependence (CD) test, CIPS panel unit root tests, panel cointegration test (first- and second-generation), fully modified ordinary least square (FMOLS) and the dynamic OLS (DOLS) and Dumitrescu and Hurlin (2012) panel causality estimation techniques. We did not provide all the estimated equations for the aforementioned econometric techniques because they are readily available in other empirical literatures. However, we focused our attention on the CS-ARDL econometric model which is our main estimation technique. Fig. 2 provides a visual representation of the methodological approach used in this study for ease of reference.

3.2. Theoretical underpinning and empirical model specification for growth and employment models

Theoretically, this paper utilizes the Cobb-Douglas (Cobb and Douglas, 1928) and Solow-Swan (Solow, 1956; Swan, 1956) production functions with necessary modification for the purpose of investigating the relationship between artificial intelligence (AI), governance quality, and economic growth within the context of BRICS economies. The aggregate growth function can be illustrated using a typical Cobb-Douglas production function with a constant rate of returns which is as follows:

$$Y_{i,t} = f\left(K_{i,t}, L_{i,t}\right) \tag{1}$$

$$Y_{i,t} = AK_{i,t}{}^{ß} L_{i,t}{}^{1-ß} \tag{2}$$

where $Y, k, LandA$ is GDP per capita (GDPPC), capital, effective labor and technical progress/efficiency, respectively. The Solow-Swan growth approach suggests that technological progress is a key driver of output. AI, as a transformative technology, can contribute to productivity gains and innovation, leading to increased output. This theory posits that AI's ability to automate tasks, optimize processes, and make predictions can enhance overall efficiency and productivity in various industries (Aghion et al., 2018; Furman and Seamans, 2019). AI can also act as a catalyst for creating new ideas and technologies, fostering long-term economic growth (Brynjolfsson and McAfee, 2014). Moreover, rapid economic expansion is associated with a wide range of factors, governance quality, economic governance, political governance, institutional governance, among others, during the production process. Given that

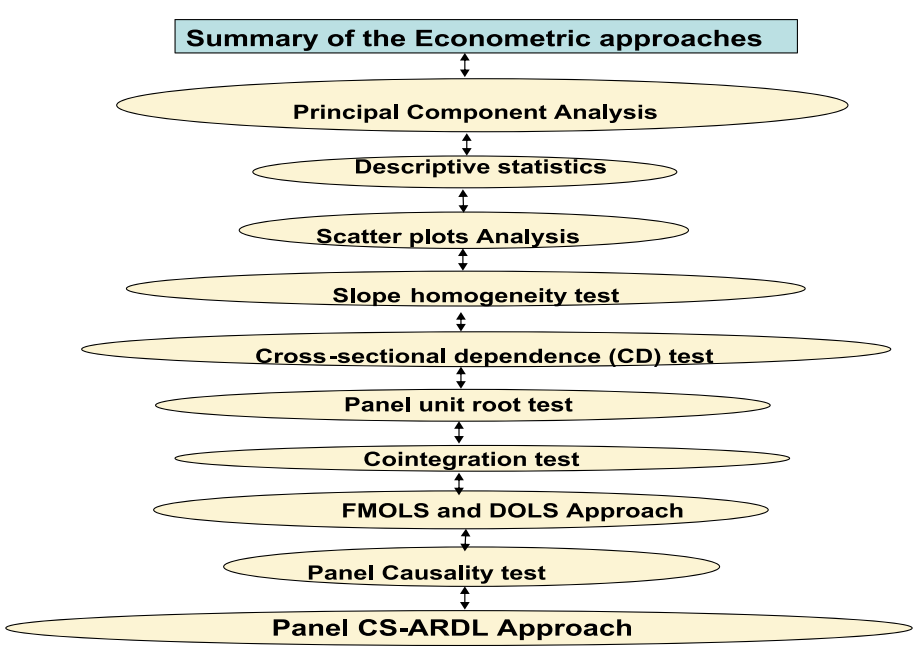

Fig. 2. Summary of the Econometric approaches.

variable A encompasses various policy-oriented variables that are not directly captured in the equation, we augment our model to include governance quality/indicators through the technical progress parameter in order to determine the empirical impact of these factors on output (Bassanini and Scarpetta, 2002; Saba, 2020). Therefore, the augmented output function can be expressed as:

$$LGDPPC_{i,t} = f\left(X_{i,t}\right) \tag{3}$$

$$LGDPPC_{i,t} = \beta_0 + \beth_0 X_{i,t} \tag{4}$$

$$LGDPPC_{i,t} = \beta_0 + \beth_0 X_{i,t} + \varepsilon_{it} \tag{5}$$

where $LGDPPC$ and X represents log of GDPPC (proxy for economic growth) and regressors,[2] respectively. For the employment model in the context of this study, economic theory suggests that technological changes in the short-term lead to a replacement effect, while in the medium to long term, the compensation theory acts to offset job losses by creating new opportunities for economic agents or members of the society (Piva and Vivarelli 2017; Freeman 1982, 1994). Compensation theory posits that advancements in technology and innovation boost productivity and wages, leading to a reduction in market prices and costs (see for example, Hicks 1963; Lorenzi and Bourlès 1995; Clark 2010; Wicksell 2022, among others). Lower costs, on one hand, enhance corporate profits and foster production expansion via investment, thereby catalyzing new job opportunities (for example, Samuelson 1988, 1989; Van Reenen 1997; Lachenmaier and Rottmann 2011, among others). On the other hand, reduced prices boost consumer purchasing power, initiating economic growth and consequently driving the generation of additional employment opportunities (see Smolny 1998; Harrison et al. 2008; Vivarelli 2014, among others). Based on the above theoretical underpinnings, the basic econometric models for employment and economic growth which were later transformed to CS-ARDL model and estimated can be found below:

Model 1:

$$\begin{aligned} LGDPPC_{i,t} = {} & \beta_1 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHMC_{i,t} \\ & + \beth_6 OGV_{i,t} + \varepsilon_{1i,t} \end{aligned} \tag{6a}$$

$$\begin{aligned} LEMPL_{i,t} = {} & \beta_2 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHMC_{i,t} \\ & + \beth_6 OGV_{i,t} + \varepsilon_{2i,t} \end{aligned} \tag{6b}$$

where β, $\beth_1, \cdots, \beth_4$, and ε_{it} represents the constants, coefficient and the error term, respectively. Model 1 excludes the interaction terms between: LAI and OGV; LAI and POLG; LAI and INSTG; LAI and ECOG; LAI*CRT; LAI*POLV; LAI*GEF; LAI*RQE; LAI*RLW; and LAI*VCAC while the rest of the models (that is, model 7a −8) does in a systemic manner one after the other.

Model 2: *Capturing the interaction between* LAI and OGV

$$\begin{aligned} LGDPPC_{i,t} = {} & \beta_1 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 OGV_{i,t} + \beth_6 LAI_{i,t} {*} OGV_{i,t} + \varepsilon_{1i,t} \end{aligned} \tag{7a}$$

$$\begin{aligned} LEMPL_{i,t} = {} & \beta_2 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 OGV_{i,t} + \beth_6 LAI_{i,t} {*} OGV_{i,t} + \varepsilon_{2i,t} \end{aligned} \tag{7b}$$

Model 3: *Capturing the interaction between* LAI and POLG

$$\begin{aligned} LGDPPC_{i,t} = {} & \beta_1 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 POLG_{i,t} + \beth_6 LAI_{i,t} {*} POLG_{i,t} + \varepsilon_{1i,t} \end{aligned} \tag{8a}$$

$$\begin{aligned} LEMPL_{i,t} = {} & \beta_2 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 POLG_{i,t} + \beth_6 LAI_{i,t} {*} POLG_{i,t} + \varepsilon_{2i,t} \end{aligned} \tag{8b}$$

Model 4: *Capturing the interaction between* LAI and ECOG

$$\begin{aligned} LGDPPC_{i,t} = {} & \beta_1 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 ECOG_{i,t} + \beth_6 LAI_{i,t} {*} ECOG_{i,t} + \varepsilon_{1i,t} \end{aligned} \tag{9a}$$

$$\begin{aligned} LEMPL_{i,t} = {} & \beta_2 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 ECOG_{i,t} + \beth_6 LAI_{i,t} {*} ECOG_{i,t} + \varepsilon_{2i,t} \end{aligned} \tag{9b}$$

Model 5: *Capturing the interaction between* LAI and INSTG

$$\begin{aligned} LGDPPC_{i,t} = {} & \beta_1 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 INSTG_{i,t} + \beth_6 LAI_{i,t} {*} INSTG_{i,t} + \varepsilon_{1i,t} \end{aligned} \tag{10a}$$

$$\begin{aligned} LEMPL_{i,t} = {} & \beta_2 + \beth_1 LAI_{i,t} + \beth_2 LGFCF_{i,t} + \beth_3 LFDI_{i,t} + \beth_4 LHUM_{i,t} \\ & + \beth_5 INSTG_{i,t} + \beth_6 LAI_{i,t} {*} INSTG_{i,t} + \varepsilon_{2i,t} \end{aligned} \tag{10b}$$

We specify the CS-ARDL model below which took it bearing from the above equations:

$$\begin{aligned} \Delta LGDPPC_{i,t} = {} & \eth_i + \xi_i\left(LGDPPC_{i,t-1} - \beth_i X_{i,t-1} - \beta_{1i}\overline{LGDPPC}_{t-1} - \beta_{2i}\overline{X}_{t-1}\right) \\ & + \sum_{j=0}^{p-1} \gamma_{i,j} \Delta LGDPPC_{i,t-j} + \sum_{j=0}^{v-1} \Gamma_{ij} \Delta X_{i,t-j} + \varnothing_{1i} \Delta \overline{LGDPPC}_t \\ & + \varnothing_{2i} \Delta \overline{X}_t + u_{it} \end{aligned} \tag{11a}$$

$$\begin{aligned} \Delta LEMPL_{i,t} = {} & \eth_i + \xi_i\left(LEMPL_{i,t-1} - \beth_i X_{i,t-1} - \beta_{1i}\overline{LEMPL}_{t-1} - \beta_{2i}\overline{X}_{t-1}\right) \\ & + \sum_{j=0}^{p-1} \gamma_{i,j} \Delta LEMPL_{i,t-j} + \sum_{j=0}^{v-1} \Gamma_{ij} \Delta X_{i,t-j} + \varnothing_{1i} \Delta \overline{LEMPL}_t \\ & + \varnothing_{2i} \Delta \overline{X}_t + u_{it} \end{aligned} \tag{11b}$$

where $\Delta LGDPPC_{i,t}$, $X_{i,t}$, $\overline{LGDPPC}_{t-1}$ & $\overline{X}_{t-1}$, $\Delta LGDPPC_{i,t-j}$&$\Delta X_{i,t-j}$, $\Delta\overline{LGDPPC}_t$&$\Delta\overline{X}_t$ and u_{it} are dependent variable, all independent variables during the long-run, mean of the dependent and explanatory variables in the long-run, dependent and independent variables in the short-run, mean dependent and independent variables during the short-run and the error term, respectively. Furthermore, where j, t, $\beth_{1i}$, γ_{1i}, Γ_{ij}, $\varnothing_{1i}$ and $\varnothing_{2i}$ denotes cross-sectional dimension, time, coefficients of the independent variables, short-run coefficient of the dependent variable, short-run coefficients of the independent variables, mean of dependent variables and mean of independent variables in the short-run, respectively (see Eq. (11a). A similar identification for variables and the parameters holds for the employment model (see Eq. (11b). The details of the dependent and independents variables can be found in Table 1. The justification for including the regressors in the models is explained briefly below:

a. *Artificial intelligence (AI):* AI influences economic growth through various ways. Firstly, AI can: *(i)* enable automation, which leads to increased productivity and efficiency in production processes (Bessen, 2018; Acemoglu, et al., 2020); *(ii)* facilitate innovation by enabling new products, services, and business models (Brynjolfsson & McAfee, 2017); *(iii)* improve decision-making through advanced data analytics, enabling better resource allocation and strategic planning (Manyika et al., 2017; Balliester and Elsheikhi, 2018). These factors collectively and ultimately can contribute to economic growth by enhancing overall productivity, competitiveness, and economic output. Furthermore, AI development and implementation create new job opportunities, for example in the areas of jobs that often involve tasks that require creativity, critical thinking, and

[2] Due to the governing rules, it is important for the reader to take note that we did not log variables with negative values.

Table 1
Variable description and data sources.

Variables	Description	Sources
Dependent variable		
LGDPPC	*Log of* GDP per capita (constant 2015 US$) serves as proxy for economic growth	WDI database
LEMPL	Employment to population ratio, 15+, total (%) (modeled ILO estimate) serves as a proxy for employment	WDI database
Independent variables		
LAI	*Log* of capital investments in artificial intelligence (AI) serves as proxy for AI	OECD database
LGFCF	*Log* of Gross fixed capital formation (% of GDP)	WDI database
LFDI	*Log* of foreign direct investment inflow (% of GDP)	WDI database
LHMC	*Log* of School enrollment, primary (% gross) serves as proxy for human capital	WDI database
LAI*OGV	Computed interaction between AI and Overall governance (OGV)	Authors
LAI* POLG	Computed interaction between AI and political governance	Authors
LAI* INSTG	Computed interaction between AI and institutional governance	Authors
LAI* ECOG	Computed interaction between AI and economic governance	Authors
LAI*CRT	Computed interaction between AI and control of corruption	Authors
LAI*POLV	Computed interaction between AI and political stability and absence of violence/terrorism	Authors
LAI*GEF	Computed interaction between AI and government effectiveness	Authors
LAI*RQE	Computed interaction between AI and regulatory quality	Authors
LAI*RLW	Computed interaction between AI and Rule of law	Authors
LAI*VCAC	Computed interaction between AI and voice and accountability	Authors
General/Overall governance (OGV) variable computed via PCA using the six governance indicators below		Authors
CRT	Control of Corruption	WGI database
POLV	Political stability and absence of violence/terrorism	WGI database
GEF	Government effectiveness	WGI database
RQE	Regulatory quality	WGI database
RLW	Rule of law	WGI database
VCAC	Voice and accountability	WGI database
POLG	It represents political governance which is computed via PCA using the voice and accountability and political stability and absence of violence/terrorism	Authors
INSTG	It represents institutional governance which is computed via PCA using the rule of law and control of corruption	Authors
ECOG	It represents economic governance which is computed via PCA using the regulatory quality and government effectiveness	Authors

Note: WDI represents World Bank's World Development Indicators. OECD represents The Organization for Economic Cooperation and Development database. WGI represents World Bank's World Governance Indicators. There were very few missing data, but this was handled by means of interpolation and extrapolation of data.
Studies that have used these techniques include those of Saba & Ngepah (2022a, 2022b, 2022c, 2022d, 2022e, 2023), Saba (2020a, 2020b, 2020c, 2021a, 2021b, 2023a, 2023b), Saba and Biyase (2022), Saba et al., (2021, 2023a, 2023b), Saba and David (2020, 2022), Saba and Ngepah (2022), Saba (2020), Saba (2021a), Saba (2020c), Saba (2020d), Saba and Ngepah (2023) and Saba et al. (2023a).

problem-solving skills, areas where human capabilities currently outshine AI (Franken and Wattenberg, 2019; Strack et al., 2021).

b. *Gross fixed capital formation (GFCF):* It refers to the total investment in fixed assets such as machinery, equipment, buildings, and infrastructure. It plays a crucial role in economic growth by increasing the quantity and quality of capital stock, and it enables businesses to produce more goods and services, leading to higher output levels/it can lead to increased productivity by providing businesses with new and more efficient capital goods (Ali, 2015). This can lead to businesses producing more output with the same amount of input, which can boost economic growth. It further boosts employment by driving economic growth and industry expansion. Increased investment in physical assets stimulates demand, creating jobs across sectors and fostering long-term stability (Megbowon et al., 2016; Pasara and Garidzirai, 2020).

c. *Foreign direct investment inflow (FDI):* It play a crucial role in fostering economic growth for both developed and developing countries given that it brings in financial resources and capital that can be used for productive purposes. This capital can fund new businesses, expand existing ones, finance infrastructure projects, creates job opportunities, among others (Mwakabungu and Kauangal, 2023). Multinational corporations that invest in foreign countries often bring advanced technologies and skills. This transfer of knowledge can have a significant impact on the host country's productivity and competitiveness (Mwakabungu and Kauangal, 2023). It is on this basis that we include the variable in our model. It enhances employment by injecting capital into the host country, fostering economic growth. Increased investment stimulates industries, creating job opportunities. Additionally, technology transfers and skill development often accompany FDI, further impacting employment positively (Megbowon et al., 2016; Osabohien et al., 2020).

d. *Human capital (HUM)*: The relationship between human capital and economic growth are well-established in the empirical literature. Human capital, which encompasses individuals' knowledge, skills, and health, plays a crucial role in driving growth (Asteriou and Agiomirgianakis, 2001). Investments in education, training, and healthcare improve the quality and productivity of the workforce, foster innovation, and enhance labor market outcomes (Barro and Lee, 2013). A highly skilled and healthy workforce contributes to increased productivity, technological advancement, and adaptability to changing economic conditions, all of which are key drivers of long-term economic growth (Barro and Lee, 2013; World Bank, 2018). Human capital, comprising skills, education, and expertise, can significantly influence employment. Well-educated and skilled individuals enhance productivity, drive innovation, and attract job opportunities. Investments in human capital lead to a more qualified and adaptable workforce, which contributes to employment outcomes (Khan and Chaudhry, 2019).

e. *Governance quality (OGV): It* simply refers to the effectiveness and efficiency of a country's governance system. A quality governance system provides a stable and transparent environment for economic activities, foster trust, protect property rights, enforce contracts, and ensure a level playing field (World Bank, 2017). Good governance, rule of law, and control of corruption can promote business confidence, attract investments, and stimulate innovation and entrepreneurship (World Bank, 2017). Sound governance system also have the capability to facilitate effective public policies and provide a conducive framework for economic growth and development. Empirical studies consistently highlight the importance of this variable to economic growth and development (see for example, Hashim et al., 2011; Fayissa and Nsiah, 2013; Singh, 2022, among others). Good governance can impact employment by fostering a stable business environment, encouraging investment, and ensuring fair labor practices. Transparent and effective governance attracts businesses, ultimately contributing to sustained employment opportunities in the long-run. (García-Castro et al., 2008; Jonasson, 2011).

3.3. Data and variables description

This research study employed annual panel data encompassing the 5-

Table 2
Principal component method results.

Panel A: Overall governance							
Principal component results							
Compnnt	Eigenvalue	Difference	Proportion	Cumulative			
Compnnt 1	3.787	2.712	0.631	0.631			
Compnnt 2	1.074	0.492	0.179	0.810			
Compnnt 3	0.582	0.248	0.097	0.907			
Compnnt 4	0.333	0.142	0.056	0.963			
Compnnt 5	0.192	0.159	0.031	0.995			
Compnnt 6	0.032		0.005	1.000			
Principal components eigenvectors results							
Variables	Compnnt 1	Compnnt 2	Compnnt 3	Compnnt 4	Compnnt 5	Compnnt 6	Unexplained
CRT	0.458	−0.043	0.097	−0.582	0.664	0.010	0.203
GEF	0.494	−0.062	0.102	−0.151	−0.482	−0.698	0.074
POLV	0.108	0.888	0.413	0.159	0.062	−0.019	0.108
RQE	0.427	−0.267	0.108	0.777	0.354	−0.072	0.232
RLW	0.489	−0.102	0.216	−0.066	−0.445	0.708	0.083
VCAC	0.336	0.352	−0.867	0.072	−0.020	0.078	0.439
Panel B: Political governance							
Compnnt	Eigenvalue	Difference	Proportion	Cumulative			
Compnnt 1	1.151	0.301	0.575	0.575			
Compnnt 2	0.849		0.425	1.000			
Principal components eigenvectors results							
Variables	Compnnt 1	Compnnt 2	Unexplained				
POLV	0.707	0.707	0.425				
VCAC	0.707	−0.707	0.425				
Panel B: Institutional governance							
Compnnt	Eigenvalue	Difference	Proportion	Cumulative			
Compnnt 1	1.903	1.807	0.952	0.952			
Compnnt 2	0.097		0.048	1.000			
Principal components eigenvectors results							
Variables	Compnnt 1	Compnnt 2	Unexplained				
CRT	0.707	0.707	0.048				
RLW	0.707	−0.707	0.048				
Panel C: Economic governance							
Compnnt	Eigenvalue	Difference	Proportion	Cumulative			
Compnnt 1	1.537	1.073	0.768	0.768			
Compnnt 2	0.463		0.232	1.000			
Principal components eigenvectors results							
Variables	Compnnt 1	Compnnt 2	Unexplained				
RQE	0.7071	0.7071	0.232				
GEF	0.7071	−0.7071	0.232				
Panel A: Correlation matrix results for the governance variable							
	i	ii	iii	iv	v	vi	
(i) CRT	1.000						
(ii) GEF	0.833***	1.000					
	(0.000)						
(iii) POLV	0.1473***	0.155***	1.000				
	(0.010)	(0.007)					
(iv) RQE	0.651***	0.749***	−0.009***	1.000			
	(0.000)	(0.000)	(0.883)				
(v) RLW	0.822***	0.962***	0.146***	0.782***	1.000		
	(0.000)	(0.000)	(0.011)	(0.000)			
(vi) VCAC	0.500***	0.549***	0.269***	0.407***	0.476***	1.000	
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)		

Note: ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$, p-value in parentheses. Where compnnt is component Source: Author's computation using WDI, WGI and ITU data.

BRICS countries, namely Brazil, Russia, India, China, and South Africa, for the period spanning from 2012 to 2022. The data sources included three primary databases, namely the Organization for Economic Cooperation and Development (OECD), the World Bank's World Development Indicators (WDI), and the World Governance Indicators (WGI). The selection of the time frame and countries was based on data availability. The variable representing governance quality was derived from the six indicators specified in Tables 1 and 2, utilizing Principal Component Analysis (PCA). Table 1 provides a comprehensive list of the variables employed in this research study.

4. Empirical results and discussion

4.1. Preliminary analysis

4.1.1. Principal component analysis

Table 2 presents the outcomes of the principal component analysis and correlation matrix for the governance quality (OGV) variable. Initially, we conducted tests to determine the extent of association among the indicators employed to construct the governance quality, political, institutional and economic governance indexes. The findings from Table 2 indicate a significant correlation between the indicators, thereby validating the prerequisite condition for performing the principal component analysis (PCA) (Saba and Ngepah, 2022a, b). To create

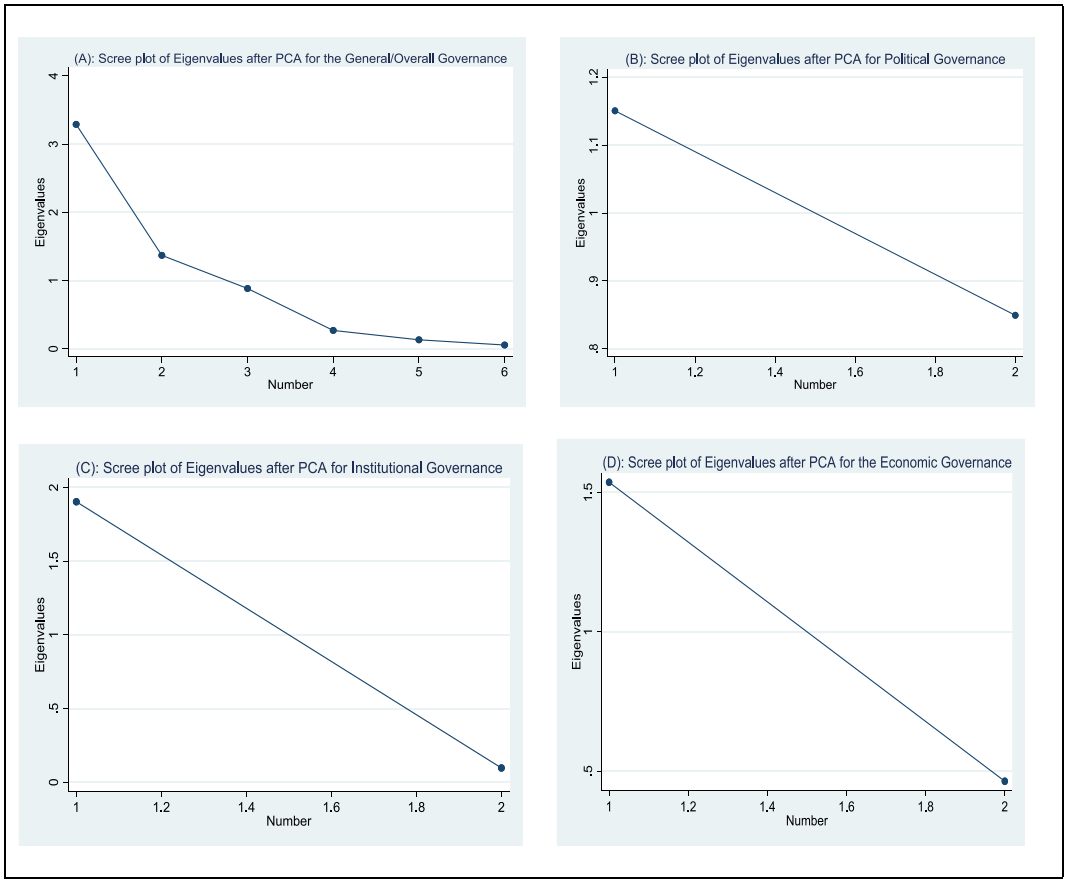

Fig. 3. (A) Scree plot for governance quality; (B) Scree plot for political governance; (C) Scree plot for institutional governance; (D) Scree plot for economic governance.

Table 3
Discriptive Statistics results.

	Mean	Median	Max	Min	Std. Dev.	Skew.	Kurtosis	Jarque-Bera	Prob	Obs.
LGDPPC	7047.190	8426.853	11189.30	1337.476	3024.169	−0.785	2.316	26.898	0.000	220
LAI	18.624	18.246	24.586	11.002	3.093	0.082	2.672	1.233	0.539	220
LFDI	0.663	0.661	2.385	−0.824	0.647	−0.090	3.636	4.013	0.134	220
LGFCF	3.168	3.064	3.796	2.572	0.369	0.394	1.947	15.857	0.000	220
LHMC	4.628	4.625	4.703	4.562	0.038	0.295	2.042	11.593	0.003	220
LEMPL	3.976	4.035	4.207	3.682	0.161	−0.246	1.589	20.474	0.000	220
OGV	1.82E-09	−0.115	5.584	−2.887	1.814	0.818	4.590	47.692	0.000	220
CRT	−0.290	−0.289	1.054	−1.052	0.437	0.857	4.705	53.575	0.000	220
GEF	0.100	0.069	1.841	−0.534	0.430	1.565	6.807	222.635	0.000	220
POLV	−0.556	−0.523	1.075	−4.270	0.656	−2.961	19.782	2903.041	0.000	220
RQE	−0.130	−0.185	0.921	−0.560	0.343	1.254	4.401	75.657	0.000	220
RLW	−0.184	−0.136	1.035	−0.870	0.407	0.444	4.096	18.236	0.000	220
VCAC	−0.207	0.295	1.112	−1.661	0.899	−0.497	1.566	27.914	0.000	220
POLG	3.27E-08	0.081	2.139	−4.338	1.073	−0.934	5.945	111.503	0.000	220
INSTG	1.27E-08	0.074	4.290	−2.401	1.380	0.603	4.333	29.630	0.000	220
ECOG	−5.45E-09	−0.075	4.546	−1.533	1.240	1.615	6.428	203.378	0.000	220

Source: Author's computations.

composite indexes for governance quality, political governance, economic governance, and institutional governance, we chose the first principal component, as it explained the highest percentage of the total variation, namely 3.79 %, 1.15 %, 1.54 % and 1.90 %, respectively. This decision was based on the eigenvalue criterion. Furthermore, Fig. 3, displaying the scree plots, provides additional support for our PCA findings.

4.1.2. Summary statistics and scatter plot analysis

Table 3 presents the summary statistics for the variables under consideration. The series with the highest mean is LGDPPC, followed by LAI and LHMC, and so on. The variables' maximum and minimum values range approximately between 11189.30 and −4.338, respectively. Skewness analysis reveals that variables with negative and positive skewness values are characterized by negatively and positively skewed distributions, respectively. Figs. 4 and 5 presents scatter plots illustrating the relationship between: (i) economic growth and the explanatory variables of interest; and employment and the explanatory variables of interest. For the economic growth model, these plots visually depict a negative association between economic growth and

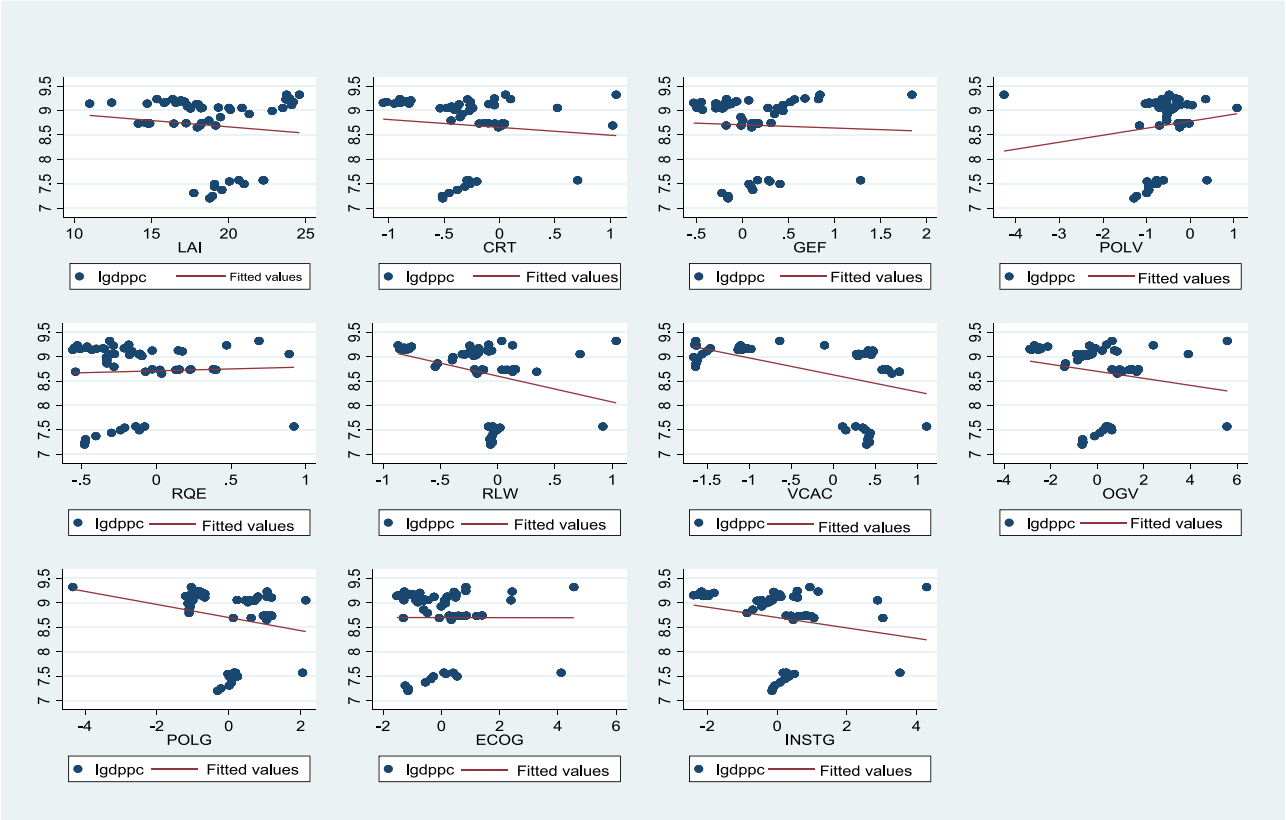

Fig. 4. Scatter plots graphical presentation between AI, governance and economic growth.

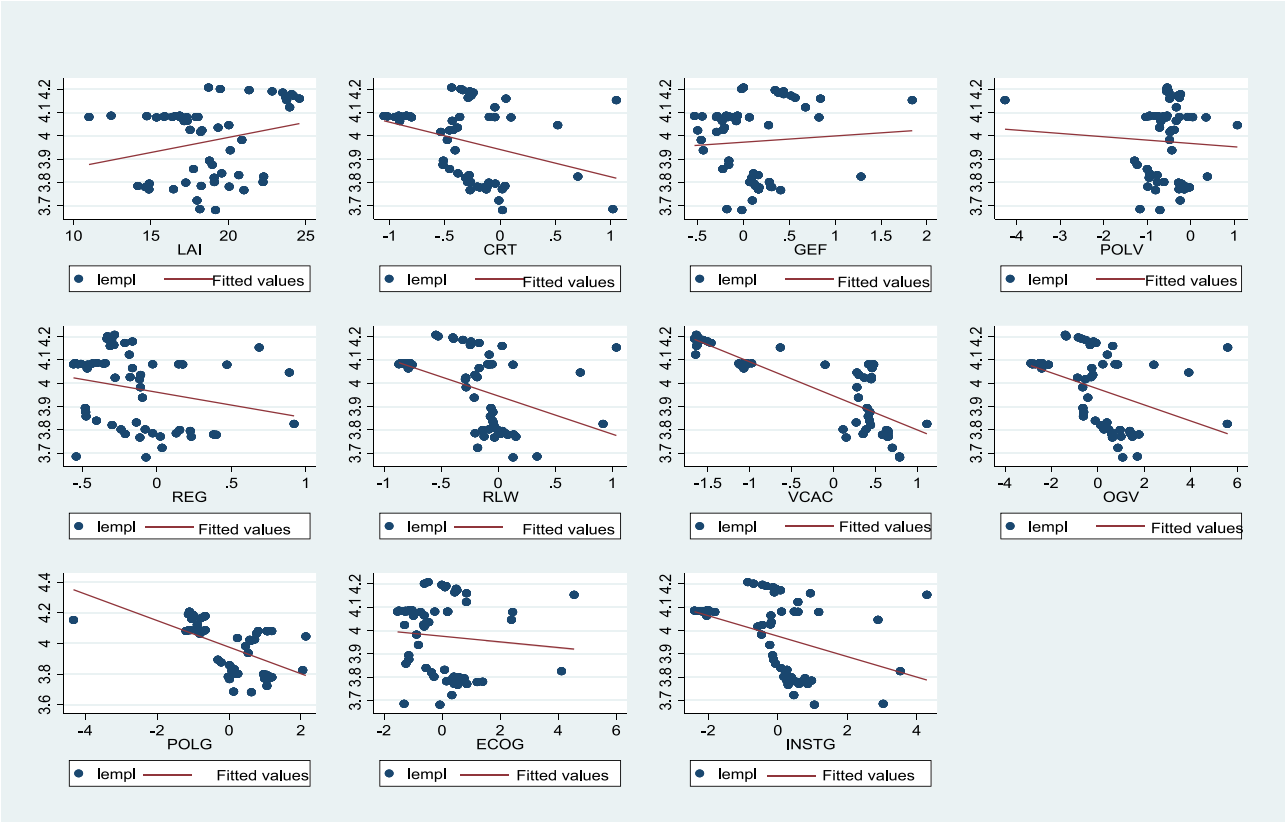

Fig. 5. Scatter plots graphical presentation between AI, governance and employment.

variables such as artificial intelligence (LAI), governance quality, government effectiveness, rule of law, voice and accountability, political governance and institutional governance while a negative scatter plot exist between economic growth and the rest of the variables. A similar interpretation also holds for the employment model. However, it is important to note that for the employment and growth models, this preliminary relationship serves only as an indication of potential associations within the examined time periods for the BRICS countries. It is crucial to highlight that the graphical approach employed in the scatter plots does not account for cross-sectional dependence issues, thereby limiting the validation of the negative/positive relationship between economic growth and the explanatory variables.

4.1.3. Slope homogeneity, cross-sectional dependence (CD), panel unit root analysis

To begin our study, we conducted a slope homogeneity test suggested by Pesaran and Yamagata (2008), and the results of the test can be found in Table 4 (for growth model) and Table 5 (for employment). Here both test statistics (delt and adjusted delta) and probability values for the two models seem to reject the assumption of homogenous slope coefficients at the 1 % significance level. This would suggest the presence of strong country heterogeneity (in the long run) for the variables under consideration. To account for cross-sectional dependence, we tested for cross-sectional dependence in the data using Pesaran (2004) and Breusch and Pagan (1980) LM test techniques. The result of the two approaches in Table 6 show that there is evidence of cross-sectional dependence in the series, with *p*-values for the statistic being statistically significant at 1 %.

Therefore, the null hypothesis (cross-sectional independence) is rejected. This implies that levels of all the variables in one member of BRICS depend on the movements of these aspects in at least one other member within the BRICS. Although the asymptotic results of all first-generation panel unit root tests (that is, Levin-Lin-Chu (LLC) (Levin et al., 2002) and Im-Pesaran-Shin (IPS) (Im et al., 2003)) rely on the assumption of cross-sectional independence, we still tested the first-generation panel unit root. However, their reliability was less important. Table 7 shows the results at levels and first differences to determine whether the variables were integrated of order zero or one. The stationarity tests for the two techniques had a null hypothesis of a unit root. While some variables were not stationary at levels, a critical examination of the results in the last column of Table 7 shows that all variables had an integrated order of 1 at a significance level of at least 1 %, indicating perfect stationarity at first differences (except for two variables).

Like the first-generation panel unit root results, the second-generation panel unit root (CIPS) results of Pesaran (2007) showed that the series were integrated of order 1 at least at a 1 % significance level (see Table 8). This indicates that we can proceed to test the long-run equilibrium relationship between the series using the second-generation cointegration approach. More importantly, based on these estimates and the aforementioned unit root and cross-dependency tests, it does confirm the adequacy of the use of the panel CS-ARDL estimator to determine the existence of a possible relationship between all the variables used in this study.

Table 4
Slope homogeneity results for the growth model.

Test statistics (Delta)	Value	p-value
Δ_{delt}	21.338***	0.000
$\Delta_{adjdelt}$	23.269***	0.000

Note: ***, ** and * denote significance at 1%, 5% and 10%, respectively. Source: Author's Computations.

Table 5
Slope homogeneity results for the employment model.

Test statistics (Delta)	Value	p-value
Δ_{delt}	17.857***	0.000
$\Delta_{adjdelt}$	19.473***	0.000

Note: ***, ** and * denote significance at 1%, 5% and 10%, respectively. Source: Author's Computations.

Table 6
Cross-sectional dependence (CD) test results.

	Pesaran test		Breusch-Pagan LM test	
Variables	Statistic	P-value	Statistic	P-value
LGDPPC	−1.02 ***	0.309	128.958***	0.000
LAI	12.42***	0.000	75.203***	0.000
LFDI	1.49**	0.036	56.247***	0.000
LGFCF	8.80***	0.000	37.116***	0.000
LHMC	−2.74***	0.006	118.860***	0.000
LEMPL	13.67***	0.000	47.917***	0.000
OGV	13.85***	0.000	180.938***	0.000
CRT	18.35***	0.000	212.046***	0.000
GEF	2.67***	0.008	176.444***	0.000
POLV	−4.26***	0.000	245.906***	0.000
RQE	4.84***	0.000	215.405***	0.000
RLW	16.27***	0.000	181.886***	0.000
VCAC	5.23***	0.000	127.863***	0.000
POLG	−3.97***	0.000	292.579***	0.000
INSTG	18.82***	0.000	223.198***	0.000
ECOG	3.82***	0.000	232.693***	0.000

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$ are significance level respectively at denote rejection of null hypothesis. Source: Author's computations.

Table 7
Panel unit root test results.

Series	Model	Levels	First Difference
LGDPPC	LLC	−1.522* (0.064)	−7.732*** (0.000)
	IPS	0.482 (0.685)	−9.015*** (0.000)
LAL	LLC	−0.970 (0.166)	−6.520*** (0.000)
	IPS	0.506 (0.694)	−7.719*** (0.000)
LFDI	LLC	0.558 (0.712)	−4.410*** (0.000)
	IPS	−1.094 (0.137)	−7.180*** (0.000)
LGFCF	LLC	−0.634 (0.263)	−6.188*** (0.000)
	IPS	−0.405 (0.343)	−7.354*** (0.000)
LHMC	LLC	0.866 (0.807)	−7.860*** (0.000)
	IPS	1.421 (0.922)	−7.387*** (0.000)
LEMPL	LLC	0.187 (0.574)	−3.21160*** (0.000)
	IPS	0.006 (0.502)	−7.350*** (0.000)
OGV	LLC	3.920 (1.000)	−9.217*** (0.000)
	IPS	2.756 (0.997)	−7.514*** (0.000)
CRT	LLC	5.074 (1.000)	−9.310*** (0.000)
	IPS	3.400 (1.000)	−7.654*** (0.000)
GEF	LLC	3.880 (1.000)	−7.576*** (0.000)
	IPS	2.937 (0.998)	−7.785*** (0.000)
POLV	LLC	4.878 (1.000)	−8.600*** (0.000)
	IPS	3.738 (1.000)	−7.772*** (0.000)
RQE	LLC	4.483 (1.000)	−8.546*** (0.000)
	IPS	3.272 (1.000)	−7.710*** (0.000)
RLW	LLC	4.405 (1.000)	−8.375*** (0.000)
	IPS	2.885 (0.998)	−7.541*** (0.000)
VCAC	LLC	2.160 (0.985)	−8.992*** (0.000)
	IPS	2.333 (0.990)	−7.790*** (0.000)
POLG	LLC	5.013 (1.000)	−8.641*** (0.000)
	IPS	3.409 (1.000)	−7.629*** (0.000)
INSTG	LLC	5.255 (1.000)	−8.800*** (0.000)
	IPS	3.440 (1.000)	−7.627*** (0.000)
ECOG	LLC	4.624 (1.000)	−8.963*** (0.000)
	IPS	3.612 (1.000)	−7.811*** (0.000)

Notes: Null: Unit root (assumes common unit root process): Levin, Lin & Chu (t*). Null: Unit root (assumes individual unit root process): Im, Pesaran and Shin (W-stat). *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$ are significance level respectively. Source: Author's computations.

Table 8
CIPS Panel unit root test results.

Variables	Levels	1st Difference
LGDPPC	−1.223	−6.190***
LAL	−1.915	−6.188***
LFDI	−1.965	−6.181***
LGFCF	−1.679	−6.181***
LHMC	−0.762	−6.187***
LEMPL	−1.709	−6.185***
OGV	−1.380	−6.187***
CRT	−1.688	−6.185***
GEF	−1.709	−6.182***
POLV	−1.303	−6.186***
RQE	−1.669	−6.185***
RLW	−1.127	−6.185***
VCAC	−0.517	−6.186***
POLG	−0.459	−6.186***
INSTG	−1.292	−6.179***
ECOG	−1.454	−6.185***

Note: *, ** and *** denote statistically significant at the 1%, 5%, and 10% level respectively. The critical values of CIPS test at 10%, 5% and 1% significance levels are: −2.21, −2.33 and −2.55 for no intercept nor trend, respectively. Source: Author's computations.

Table 9
Johansen-Fisher Panel cointegration test results for the growth model.

		Trace test				Maximum Eigenvalue test	
H_0	H_1	λ-trace statistic	p-value	H_o	H_1	λ-max statistic	p-value
$r = 0$	$r \geq 1$	0.000	1.000	$r = 0$	$r \geq 1$	0.000	1.000
$r \leq 1$	$r \geq 2$	107.0***	0.000	$r \leq 1$	$r \geq 2$	83.46***	0.000
$r \leq 2$	$r \geq 3$	245.7***	0.000	$r \leq 2$	$r \geq 3$	217.6***	0.000
$r = 3$	$r \geq 4$	115.1***	0.000	$r = 3$	$r \geq 4$	93.81***	0.000
$r \leq 4$	$r \geq 5$	57.84***	0.000	$r \leq 4$	$r \geq 5$	45.44***	0.000
$r \leq 5$	$r \geq 6$	34.21***	0.000	$r \leq 5$	$r \geq 6$	34.21***	0.000

Notes: *Rejection of the null hypothesis of no cointegration at least at the 10% level of significance. Probabilities are computed using asymptotic Chi-square distribution. Source: Author's computations.

4.2. The cointegration analysis

The Johansen-Fisher (see Tables 9 and 10) and Westerlund (2007) panel cointegration tests (see Tables 11 and 12) which were conducted to establish the existence of long-run equilibrium relationships between the dependent variables (that is, the employment and economic growth) and the respective regressors. Before estimating the two approaches, we first determined the optimum lag length to be used for the estimations. The results in Table 13 and 14 indicate that the optimum lag length is 1, following the results of the three information criteria.

Table 10
Johansen-Fisher Panel cointegration test results for the employment model.

		Trace test				Maximum Eigenvalue test	
H_0	H_1	λ-trace statistic	p-value	H_o	H_1	λ-max statistic	p-value
$r = 0$	$r \geq 1$	46.331***	0.004	$r = 0$	$r \geq 1$	17.842***	0.001
$r \leq 1$	$r \geq 2$	28.489***	0.008	$r \leq 1$	$r \geq 2$	14.550***	0.006
$r \leq 2$	$r \geq 3$	13.939***	0.005	$r \leq 2$	$r \geq 3$	11.348***	0.001
$r = 3$	$r \geq 4$	2.591***	0.000	$r = 3$	$r \geq 4$	1.997***	0.000
$r \leq 4$	$r \geq 5$	0.594***	0.010	$r \leq 4$	$r \geq 5$	0.591***	0.009
$r \leq 5$	$r \geq 6$	0.004***	0.005	$r \leq 5$	$r \geq 6$	0.004***	0.005

Notes: *Rejection of the null hypothesis of no cointegration at least at the 10% level of significance. Probabilities are computed using asymptotic Chi-square distribution. Source: Author's computations.

Table 11
Westerlund panel cointegration tests for the growth model.

Statistic	Value	Z-value	P-value	Robust P-value
G_t	−1.846**	1.877	0.170	0.020
G_a	−6.541**	2.288	0.189	0.030
P_t	−4.322*	0.910	0.219	0.070
P_a	−6.987*	1.161	0.177	0.070

Note: *, ** and *** represent significance at the 1%, 5%, and 10% levels respectively; number of replications to obtain bootstrapped p-values is set to 100; bandwidth is selected according to the data depending rule $4(\frac{T}{100})^{2/9} \approx 3$ recommended by Newey and West (1994); Barlett is used as the spectral estimation method. Source: Author's Computations.

For the growth model, in Tables 9, the Johansen-Fisher panel cointegration results reveal ten cointegrating vectors, five from the trace statistic and five from the maximum eigenvalue statistic. This indicates the existence of a long-run equilibrium relationship between the variables used in this study. We rejected the null hypothesis of no cointegration at a significance level of at least 1 % for both the trace and maximum eigenvalue statistic values, thereby concluding that there is evidence of cointegration. A similar interpretation also applies to the employment model; however, in this case, the results indicate twelve cointegrating vectors, with six from the trace statistic and six from the maximum eigenvalue statistic (see Table 10).

For the growth model, in Tables 11, For the purpose of robustness, we employed the Westerlund (2007) panel cointegration test to address the issue of cross-sectional dependence among countries. This test is considered more reliable than older ones (Khan et al., 2020). Results from the test in Table 11 indicate that all the four tests (that is the values of G_t, G_a, P_t and P_a statistics) has a probability value smaller than 10 % by considering the robust p-value. However, the rejection of the null hypothesis of no-cointegration for at least one test implies the existence of a long-run equilibrium relationship between the variables. This suggests that even with interconnections among countries, there is still a long-run relationship between the variables. For the employment model in Tables 12, a similar conclusion can be drawn from the results, as all four tests have a probability value smaller than 10 %, considering the robust p-value. This suggests a long-run equilibrium relationship between employment and other explanatory variables in the employment model.

4.3. Fully modified least squares (FMOLS) and dynamic OLS (DOLS) long run analysis

We applied the FMOLS and DOLS techniques proposed by Pedroni (2001, 2004) to estimate the long-run coefficients of explanatory variables in the growth and employment models (refer to Table 15 and 16). These techniques address serial correlation and endogeneity issues more effectively than OLS. For the both models, the R-squared and Adjusted R-

Table 12
Westerlund panel cointegration tests for the employment model.

Statistic	Value	Z-value	P-value	Robust P-value
G_t	−1.799*	1.987	0.077	0.010
G_a	−7.040*	2.153	0.084	0.080
P_t	−4.483*	0.762	0.077	0.010
P_a	−8.361*	0.793	0.786	0.080

Note: *, ** and *** represent significance at the 1%, 5%, and 10% levels respectively; number of replications to obtain bootstrapped p-values is set to 100; bandwidth is selected according to the data depending rule $4(\frac{T}{100})^{2/9} \approx 3$ recommended by Newey and West (1994); Barlett is used as the spectral estimation method. Source: Author's Computations.

Table 13
Optimum lag length selection results for the economic growth model.

Lag	AIC	SIC	HQIC
0	9.072	9.172	9.113
1	−11.914	−11.209*	−11.629*
2	−11.597	−10.288	−11.067
3	−11.306	−9.393	−10.531
4	−11.078	−8.560	−10.058
5	−12.140*	−9.018	−10.876

Note: * indicates lag order selected by the criterion. AIC is Akaike information criterion; SIC is Schwarz information criterion;
Hannan-Quinn information criterion. Source: Author's computations.

squared values, which were above 50 % for both approaches, demonstrate that all our models were correctly specified. Therefore, we can proceed to interpret and discuss our estimated results. In Table 15 and 16, Panels A and B present the results of the FMOLS and DOLS, respectively. The impact of the explanatory variables on economic growth (see Table 15) and employment (refer to 16) appears to be similar in terms of sign, coefficient, and significance for both FMOLS and DOLS models.

Therefore, we will focus on interpreting and discussing our main variables of interesting to save space. In Panel A and B of Table 15, in the long run, especially for the results in Column 2, artificial intelligence (AI) and overall governance quality (OGV) significantly and positively impact economic growth for the BRICS countries for both approaches. This aligns with the research conducted by Aghion et al. (2018), Lu (2021), and Acemoglu and Restrepo (2018b) regarding the AI-growth nexus. It also concurs with the research conducted by Kaufmann et al. (2000), Knack (2002), Kraay et al. (2003), Huang and Ho (2017), Raza et al. (2021), Singh (2022), and others focusing on the governance-growth nexus. A 1 % increase in AI and OGV increased economic growth by 0.024 % and 0.030 % for the FMOLS model and by 0.015 % and 0.155 % for the DOLS model. In Column 3 of Table 15, the interaction between AI and governance quality has a significant and positive impact on economic growth for both the FMOLS and DOLS models. A 1 % increase in the interaction between AI and governance quality increased economic growth by 0.002 % and 0.011 % for the FMOLS and DOLS models, respectively. This implies that the interaction between AI and governance quality plays a significant role in promoting growth in BRICS countries.

While in Panel A and B of Table 16, in the long run, especially for the results in Column 2, artificial intelligence (AI) and overall governance quality (OGV) significantly and negatively impact employment for the BRICS countries for both approaches. This aligns with the research conducted by Baruch and Rosenstein (1992), Greenhaus and Kossek (2014), Acemoglu and Restrepo, (2018c), Yang and Fan (2020), Agrawal et al. (2023), among others. A 1 % increase in AI and OGV reduced employment by 0.003 % and 0.038 % for the FMOLS model and by 0.001 % and 0.052 % for the DOLS model. In Column 3 of Table 16, the interaction between AI and governance quality has a significant and positive impact on employment for both the FMOLS and DOLS models. A 1 % increase in the interaction between AI and governance quality increased employment by 0.003 % and 0.006 % for the FMOLS and DOLS models, respectively. This implies that the interaction between AI and governance quality will play a significant role in promoting employment in BRICS countries in the long run.

Table 14
Optimum lag length selection results for the employment model.

Lag	AIC	SIC	HQIC
0	5.434	5.535	5.475
1	−12.653	−11.948*	−12.368*
2	−12.305	−10.996	−11.775
3	−11.964	−10.051	−11.190
4	−11.643	−9.126	−10.624
5	−12.730*	−9.608	−11.466

Note: * indicates lag order selected by the criterion. AIC is Akaike information criterion; SIC is Schwarz information criterion;
Hannan-Quinn information criterion. Source: Author's computations.

In Column 4 of Table 15, the interaction between AI and political governance (POLG) has a significant and negative impact on growth for both the FMOLS and DOLS models. A 1 % increase in the interaction between AI and political governance decreased growth by 0.053 % and 0.013 % for the FMOLS and DOLS models, respectively. This implies that the interaction between AI and political governance has not played a significant role in promoting growth in BRICS countries. In Column 5 of Table 15, the interaction between AI and institutional governance (INSTG) has a significant and positive impact on growth for both the FMOLS and DOLS models. A 1 % increase in the interaction between AI and institutional governance increased growth by 0.004 % and 0.006 % for the FMOLS and DOLS models, respectively. This implies that the interaction between AI and institutional governance played a significant role in promoting growth in BRICS countries. In Column 6 of Table 15, a similar interpretation also holds for the interaction between AI and economic governance (ECOG). This implies that the interaction between AI and economic governance played a significant role in promoting growth in BRICS countries.

In Column 4, 5 and 6 of Table 16, the interaction between AI and governance dimensions such as political governance (POLG), institutional governance (INSTG) and economic governance (ECOG) has a significant and positive impact on employment for both the FMOLS and DOLS models. This implies that the interaction between AI and the mentioned variables (that is, political, institutional and economic governance) played a significant role in promoting employment in BRICS countries. In Column 7, 9, 10 and 11 of Table 16, the interaction between AI and governance indicators such as control of corruption (CRT), government effectiveness (GEF), regulatory quality (RQE) and rule of law (RLW) has a significant and positive impact on employment for both the FMOLS and DOLS models. This implies that the interaction between AI and explanatory variables such as CRT, GEF, RQE and RLW played a significant role in promoting employment in BRICS countries. While the opposite results holds when AI is interacted with political stability (POLV) and voice and accountability (VCAC).

In Column 7, 9, 10 and 11 of Table 15, the interaction between AI and explanatory variables such as control of corruption (CRT), government effectiveness (GEF), regulatory quality (RQE) and rule of law (RLW) has a significant and positive impact on growth for both the FMOLS and DOLS models. A 1 % increase in the interaction between AI and explanatory variables such as CRT, GEF, RQE and RLW increased growth by 0.018 %, 0.019 %, 0.091 % and 0.012 % for the FMOLS model, and by 0.021 %, 0.025 %, 0.083 % and 0.019 % for the DOLS models, respectively. This implies that the interaction between AI and explanatory variables such as CRT, GEF, RQE and RLW played a significant role in promoting growth in BRICS countries. In Column 8 and 12 of Table 15, the interaction between AI and explanatory variables such as political stability (POLV) and voice and accountability (VCAC) has a significant and negative impact on growth for both the FMOLS and DOLS models. A 1 % increase in the interaction between AI and explanatory variables such as POLV and VCAC decreased growth by 0.014 % and 0.010 % for the for the FMOLS model, and by 0.192 % and 0.012 % for the DOLS models, respectively. This implies that the interaction between AI and explanatory variables such as POLV and VCAC has not played a significant role in promoting growth in BRICS countries.

4.4. Panel causality analysis

This section analysed the causal relationship between the series under review. We rejected/accepted the null hypothesis that there was no causation for each Chi square-value statistic since their p-values were less/greater than the 10 % significance level. Table 17 presents the panel

Table 15
FMOLS and DOLS estimates for the economic growth model.

PANEL A: FMOLS											
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff
LAI	0.024***	0.024***	0.015	0.026***	0.023***	0.032***	0.022***	0.019***	0.026***	0.0277***	0.026***
	(0.001)	(0.001)	(0.023)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
LFDI	−0.040**	−0.037***	0.128***	−0.039***	−0.026***	−0.036***	−0.030***	−0.024***	−0.027***	−0.039***	−0.030***
	(0.003)	(0.002)	(0.012)	(0.003)	(0.002)	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)	(0.003)
LGFCF	−0.082***	−0.053***	−1.617***	−0.012	−0.058***	0.014***	−0.014	0.007	−0.099***	−0.033*	0.029*
	(0.021)	(0.017)	(0.001)	(0.019)	(0.017)	(0.019)	(0.017)	(0.019)	(0.020)	(0.019)	(0.017)
LHUM	0.449***	0.544***	3.097***	0.019***	0.500***	0.784***	0.5409***	0.202***	0.710***	0.565***	0.457***
	(0.069)	(0.053)	(0.001)	(0.061)	(0.054)	(0.059)	(0.056)	(0.064)	(0.058)	(0.061)	(0.063)
OGV	0.030***	−0.007***									
	(0.007)	(0.005)									
LAI*OGV		0.002***									
		(0.000)									
POLG			1.334***								
			(0.000)								
LAI*POLG			−0.053***								
			(0.001)								
INSTG				−0.053***							
				(0.009)							
LAI* INSTG				0.004***							
				(0.000)							
ECOG					−0.035***						
					(0.007)						
LAI* ECOG					0.004***						
					(0.000)						
CRT						−0.256***					
						(0.027)					
LAI*CRT						0.018***					
						(0.001)					
POLV							0.284***				
							(0.018)				
LAI*POLV							−0.014***				
							(0.001)				
GEF								−0.249***			
								(0.027)			
LAI*GEF								0.019***			
								(0.001)			
RQE									0.082***		
									(0.029)		
LAI*RQE									0.091***		
									(0.010)		
RLW										−0.117***	
										(0.030)	
LAI*RLW										0.012***	
										(0.001)	
VCAC											0.289***
											(0.019)
LAI*VCAC											−0.010***
											(0.0010)
R-squared	0.988	0.99	0.77	0.99	0.99	0.99	0.99	0.99	0.98	0.98	0.98
Adj. R-squared	0.987	0.99	0.77	0.99	0.99	0.99	0.99	0.99	0.98	0.98	0.98
Obs	215	215	215	215	215	215	215	215	215	215	215
PANEL B: DOLS											
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff
LAI	0.015***	0.003	0.027***	0.016***	0.011**	0.022***	−0.101***	0.009**	0.004	0.019***	0.013**
	(0.005)	(0.006)	(0.007)	(0.005)	(0.004)	(0.006)	(0.016	(0.003)	(0.007)	(0.005)	(0.006)
LFDI	−0.043**	−0.061***	−0.029	−0.011	−0.001	−0.009	0.073***	0.001	0.030**	−0.013	−0.009
	(0.018)	(0.019)	(0.020)	(0.013)	(0.011)	(0.013)	(0.024)	(0.009)	(0.015)	(0.012)	(0.012)
LGFCF	−0.272***	−0.286**	−0.035	0.114	0.053	0.121	0.720	0.089	0.298***	0.107	0.140*
	(0.101)	(0.101)	(0.146)	(0.098)	(0.094)	(0.096)	(0.143)	(0.073)	(0.071)	(0.100)	(0.084)
LHMC	−0.672**	−2.111***	0.130	0.991***	0.716**	1.100***	1.880***	0.630**	1.706***	0.866**	0.455
	(0.308)	(0.621)	(0.519)	(0.327)	(0.303)	(0.313)	(0.084)	(0.257)	(0.049)	(0.334)	(0.408)
OGV	0.155***	0.314***									
	(0.017)	(0.077)									
LAI*OGV		0.011**									
		(0.005)									
POLG			0.261**								
			(0.106)								
LAI*POLG			−0.013**								
			(0.005)								
INSTG				−0.092**							
				(0.045)							

(*continued on next page*)

Table 15 (*continued*)

PANEL A: FMOLS (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
LAI*INSTG				0.006***							
				(0.002)							
ECOG					−0.079**						
					(0.038)						
LAI* ECOG					0.006***						
					(0.002)						
CRT						−0.346***					
						(0.132)					
LAI*CRT						0.021***					
						(0.007)					
POLV							4.617***				
							(0.474)				
LAI*POLV							−0.192***				
							(0.022)				
GEF								−0.362***			
								(0.100)			
LAI*GEF								0.025***			
								(0.005)			
RQE									−1.532***		
									(0.354)		
LAI*RQE									0.083***		
									(0.020)		
RLW										−0.265*	
										(0.143)	
LAI*RLW										0.019***	
										(0.007)	
VCAC											0.324**
											(0.134)
LAI*VCAC											−0.012**
											(0.006)
R-squared	0.99	0.99	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.98
Adjusted R-squared	0.997	0.99	0.98	0.98	0.98	0.99	0.99	0.98	0.99	0.99	0.98
Observations	200	200	215	220	220	220	220	220	220	220	220

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard Error in Parenthesis, while the dependent variable is real GDP per capita which serves as a proxy for economic growth. Source: Author's Computations.

causality test results. In Table 17, focusing on our variables of interest, unidirectional causality runs from: (i) growth to AI; (ii) governance quality to AI; (ii) political governance to AI; (iii) institutional governance to AI; (iv) economic governance to AI; (v) AI to employment; and (vi) the six governance indicators to AI (except voice and accountability). This implies that investment in AI is dependent on growth, governance quality, political governance, institutional governance, economic governance and the six governance indicators (except voice and accountability).

The causality results suggest several important implications: *(i)* the unidirectional causality running from AI to employment suggests that as AI technology advances and becomes more integrated into various sectors, it can directly affect job creation or modification in these economies. This causality outcome indicates that the progress in AI technology is a key driver in shaping employment trends and structures in BRICS nations, possibly leading to new job opportunities or the transformation of existing ones; *(ii)* the unidirectional causality from growth to AI implies that growth can be a driver for the adoption and development of artificial intelligence. As BRICS economies grow, they typically have more resources available for research and development, education, and technology investment.

This result suggests that as BRICS countries experience economic growth, they are more likely to invest in AI technologies; *(iii)* the causality from governance quality to AI implies that better governance practices within a country can facilitate the development and implementation of AI. Good governance often involves transparent and efficient regulatory frameworks, which can create an environment conducive to technological innovation. This suggests that improvements in governance quality can support AI adoption; *(vi)* the unidirectional causality from political governance to AI suggests that the political stability and effectiveness of a country's government can impact its AI development.

Stable political environments are more likely to provide consistent support for long-term technological projects, including AI research and development in BRICS; (v) the causality from institutional governance to AI indicates that strong institutions, including legal systems and property rights protection, are important for AI development. These institutions provide a stable and secure environment for innovation and investment in AI technologies; *(vi)* the causality from economic governance to AI implies that factors like ease of doing business, trade policies, and competition regulations can influence a country's AI development. Favorable economic governance can attract foreign investment and foster domestic entrepreneurship in the AI sector; *(vii)* the causality from most governance indicators to AI (except voice and accountability) suggests that various aspects of governance, such as control of corruption, government effectiveness, regulatory quality, and rule of law, play a role in AI development.

These factors collectively contribute to creating an environment that is conducive to technological advancement. The results further suggest that robust governance practices, political stability, and supportive economic policies for innovation can facilitate the growth of AI. Moreover, economic growth itself can stimulate investment in AI technologies. These causality findings underscore the significance of adopting a comprehensive approach to promote AI development, taking into account economic, political, and governance dimensions. Fig. 6 provides the summary of the panel causality results.

4.5. Panel CS-ARDL analysis

Table 18 presents the panel CS-ARDL results. Commencing with the

Table 16
FMOLS and DOLS estimates for the employment model.

PANEL A: FMOLS											
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff
LAI	−0.003***	0.003	−0.005***	0.003***	0.002**	0.005***	−0.003***	0.002**	0.003***	0.004***	−0.008***
	(0.000)	(0.007)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
LFDI	0.052***	0.056***	0.037***	0.062***	−0.014***	−0.016***	−0.013***	−0.014***	−0.014***	−0.015***	−0.007***
	(0.003)	(0.022)	(0.002)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
LGFCF	0.261***	0.213***	0.169***	0.225***	0.080***	0.086***	0.103***	0.096***	0.078***	0.092***	0.169***
	(0.007)	(0.052)	(0.006)	(0.007)	(0.016)	(0.017)	(0.016)	(0.013)	(0.017)	(0.014)	(0.014)
LHUM	0.684***	0.692***	0.758***	0.680***	0.219***	0.227***	0.283***	0.223***	0.216***	0.207***	0.061***
	(0.003)	(0.025)	(0.003)	(0.003)	(0.057)	(0.054)	(0.059)	(0.051)	(0.061)	(0.046)	(0.050)
OGV	−0.038***	−0.137***									
	(0.001)	(0.039)									
LAI*OGV		0.003**									
		(0.002)									
POLG			−0.181***								
			(0.008)								
LAI*POLG			0.006***								
			(0.000)								
INSTG				−0.201***							
				(0.007)							
LAI* INSTG				0.008***							
				(0.000)							
ECOG					−0.011						
					(0.007)						
LAI* ECOG					0.001***						
					(0.000)						
CRT						−0.059***					
						(0.023)					
LAI*CRT						0.004***					
						(0.001)					
POLV							0.089***				
							(0.018)				
LAI*POLV							−0.004***				
							(0.001)				
GEF								−0.045*			
								(0.025)			
LAI*GEF								0.004***			
								(0.001)			
RQE									−0.044		
									(0.027)		
LAI*RQE									0.004***		
									(0.001)		
RLW										−0.060***	
										(0.021)	
LAI*RLW										0.004***	
										(0.001)	
VCAC											0.037***
											(0.015)
LAI*VCAC											−0.001***
											(0.001)
R-squared	0.56	0.59	0.54	0.63	0.99	0.99	0.99	0.98	0.98	0.98	0.98
Adj. R-squared	0.55	0.58	0.53	0.62	0.98	0.98	0.98	0.98	0.98	0.98	0.97
Obs	215	215	215	215	215	215	215	215	215	215	215
PANEL B: DOLS											
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff	Coeff
LAI	−0.001***	0.003	0.001	0.006	−0.002	0.015**	−0.005	−0.003	−0.001	0.005	−0.008*
	(0.000)	(0.006)	(0.005)	(0.004)	(0.005)	(0.007)	(0.010)	(0.004)	(0.006)	(0.005)	(0.005)
LFDI	0.058***	0.038*	0.022	0.038***	0.032***	0.042**	0.059*	0.045***	0.036*	0.039***	0.042***
	(0.026)	(0.023)	(0.019)	(0.014)	(0.019)	(0.018)	(0.031)	(0.016)	(0.020)	(0.016)	(0.013)
LGFCF	0.214***	0.187***	0.190***	0.252***	0.301***	0.227***	0.283***	0.305***	0.307***	0.292***	0.079**
	(0.057)	(0.054)	(0.059)	(0.041)	(0.041)	(0.048)	(0.074)	(0.034)	(0.046)	(0.043)	(0.035)
LHMC	0.705***	0.709***	0.730***	0.657***	0.659***	0.628***	0.675***	0.655***	0.653***	0.623***	0.823***
	(0.027)	(0.026)	(0.034)	(0.021)	(0.022)	(0.023)	(0.040)	(0.021)	(0.023)	(0.020)	(0.020)
OGV	−0.052***	−0.155***									
	(0.009)	(0.036)									
LAI*OGV		0.006***									
		(0.002)									
POLG			−0.266***								
			(0.058)								
LAI*POLG			0.010***								
			(0.003)								

(*continued on next page*)

Table 16 (*continued*)

PANEL A: FMOLS (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
INSTG				−0.196***							
				(0.036)							
LAI*INSTG				0.006***							
				(0.002)							
ECOG					−0.247***						
					(0.050)						
LAI* ECOG					0.010***						
					(0.003)						
CRT						−0.656***					
						(0.147)					
LAI*CRT						0.026***					
						(0.008)					
POLV							−0.334*				
							(0.179)				
LAI*POLV							0.016**				
							(0.008)				
GEF								−1.078***			
								(0.159)			
LAI*GEF								0.045***			
								(0.008)			
RQE									−0.836***		
									(0.204)		
LAI*RQE									0.039***		
									(0.012)		
RLW										−0.575***	
										(0.125)	
LAI*RLW										0.021***	
										(0.007)	
VCAC											−0.120**
											(0.061)
LAI*VCAC											−0.001
											(0.003)
R-squared	0.79	0.84	0.77	0.82	0.77	0.79	0.74	0.79	0.74	0.83	0.84
Adjusted R-squared	0.66	0.70	0.58	0.66	0.57	0.61	0.51	0.60	0.51	0.68	0.81
Observations	205	205	205	205	205	205	205	205	205	205	205

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard Error in Parenthesis, while the dependent variable is employment. Source: Author's Computations.

Table 17
Dumitrescu and Hurlin (2012) panel causality test results.

Model	Null hypothesis	W-statistic	Zbar-statistic	p-value	Direction of relationship observed	Conclusion
1	LGDPPC ↛ LAI	1.268**	−1.527	0.027	LGDPPC → LAI	Unidirectional causality
	LAI ↛ LGDPPC	2.322	−0.689	0.491		
2	LAI ↛ LEMPL	1.795**	−2.389	0.017	LAI → LEMPL	Unidirectional causality
	LEMPL ↛ LAI	7.478	0.483	0.629		
3	OGV ↛ LAI	0.381**	−2.233	0.026	OGV → LAI	Unidirectional causality
	LAI ↛ OGV	3.323	0.107	0.914		
4	POLG ↛ LAI	0.402**	−2.215	0.027	POLG → LAI	Unidirectional causality
	LAI ↛ POLG	2.992	−0.155	0.877		
5	INSTG ↛ LAI	0.216**	−2.364	0.018	INSTG → LAI	Unidirectional causality
	LAI ↛ INSTG	3.879	0.550	0.582		
6	ECOG ↛ LAI	0.561**	−2.090	0.037	ECOG → LAI	Unidirectional causality
	LAI ↛ ECOG	2.624	−0.448	0.654		
7	VCAC ↛ LAI	2.477	−0.565	0.572	VCAC ↛ LAI	No causality
	LAI ↛ VCAC	1.137	−1.631	0.103		
8	RLW ↛ LAI	0.390**	−2.226	0.026	RLW → LAI	Unidirectional causality
	LAI ↛ RLW	3.445	0.205	0.838		
9	RQE ↛ LAI	0.750**	−1.939	0.053	RQE → LAI	Unidirectional causality
	LAI ↛ RQE	3.592	0.322	0.747		
10	POLV ↛ LAI	0.578**	−2.076	0.038	POLV → LAI	Unidirectional causality
	LAI ↛ POLV	3.177	−0.008	0.993		
11	GEF ↛ LAI	0.546**	−2.101	0.036	GEF → LAI	Unidirectional causality
	LAI ↛ GEF	1.853	−1.061	0.289		
12	CRT ↛ LAI	0.470**	−2.161	0.031	CRT → LAI	Unidirectional causality
	LAI ↛ CRT	3.305	0.094	0.926		

Note: ↔ and → denote bidirectional and unidirectional causality respectively. ↛ denote does not homogeneously cause (i.e H_0). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Source: Author's Computations.

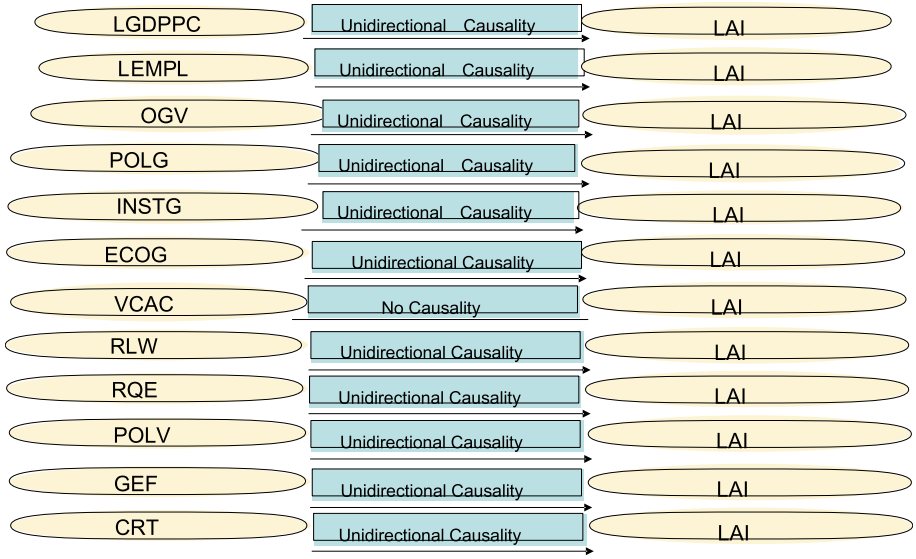

Fig. 6. Summary of the Casuality Results.

estimates of the Error Correction Term (ECT) for all regression models, the values are −1.076, −1.004, −1.125, −1.030, −1.132, −1.061, −1.015, −1.113, −1.290, −1.063, and −1.086 in Column 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11, respectively, all of which are statistically significant, at least at the 1 % level of significance for the economic growth model. For the employment model, in Table 19, the values are −0.996, −1.012, −1.007, −1.015, −1.055, −1.059, −1.098, −1.040, −1.074, −0.998 and −1.342 in Column 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and 11, respectively, all of which are statistically significant, at least at the 1 % level of significance. These values suggest that there is a strong negative relationship between the deviations from the long-run equilibrium and the short-run changes in the economic growth and employment variables. Specifically, the negative ECT values indicate that any deviations from the long-run equilibrium will be corrected at a rate of those values, suggesting that the economic growth and employment variables will adjust back towards its equilibrium level relatively quickly. Additionally, the R-squared values for all the models are above 50 %, which implies that our models are correctly specified.

Focusing on our variables of interest, for the first model which shows the results without the interaction term variables, in Column 1 of Table 18, the CS-ARDL results show that both in the short- and long-run, at a 10 % level of significance, the impact of AI and governance quality on growth is significant and positive. Specifically, a 1 % increase in AI and governance quality increased short-run growth by 0.001 % and 0.042 %, and long-run growth by 0.001 % and 0.039 %, respectively. In Table 19, a 1 % increase in AI and governance quality increased short-run employment by 0.002 % and 0.001 %, and long-run employment by 0.002 % and 0.001 %, respectively. For growth model, the results suggest that in the short-run, the impact of AI and governance quality on growth is more pronounced (0.001 % for AI and 0.042 % for governance quality).

This aligns with the research conducted by Lu (2021), who explores the impact of AI on growth and welfare. The findings indicate that AI advancements have the potential to boost growth in the transitional dynamics phase and can lead to short-term utility gains for households, especially when AI accumulation results from increased productivity in the goods sector or the AI sector. However, in the long run, these effects become less pronounced, particularly for governance quality, which is 0.039 %. This implies that investments in AI and improvements in governance deliver more substantial short-term benefits, and these benefits may diminish in their long-term impact on growth for BRICS countries.

For the employment model, the result suggests that in both short- and long-run, the impact of AI and governance quality on employment appears to be similar in co-efficient. These results imply that improvements in AI and governance quality have a positive, though modest, impact on employment. Specifically, for every 1 % increase in AI and governance quality, there's a corresponding slight increase in both short-term and long-term employment. This suggests that advancements in AI and better governance can contribute to employment growth, albeit at a relatively small scale. This aligns with the research conducted by Guliyev (2023), but contradicts the findings of Leontief (1983), Ngo et al. (2014), Susskind and Susskind (2016), Frey and Osborne (2017), David (2017), among others.

In both the short- and long-run, the consistent positive and significant impact of the unconditional variables such as POLV and VCAC on economic growth aligns with previous findings in studies by Jalilian et al. (2006), Huynh and Jacho-Chávez (2009), Gani (2011), Méndez-Picazo et al. (2012), and others. In both the short- and long-run, the consistent positive and significant impact of the unconditional variables such as POLV and VCAC on economic growth aligns with previous findings in studies by Jalilian et al. (2006), Huynh and Jacho-Chávez (2009), Gani (2011), Méndez-Picazo et al. (2012), and others. The results for CRT, GEF, and RLW are consistent with the findings of Mo (2001), Méon and Sekkat (2005), Venard (2013), Ugur and Dasgupta (2011), among others, while it contradicts the findings of Dollar and Kraay (2000), Butkiewicz and Yanikkaya (2006), Alam et al. (2017), among others.

In Column 2, 3, and 4 of Table 18, the results indicate that the interaction between AI and explanatory variables such as overall governance, political governance, and institutional governance has a negative and significant impact on growth at a 10 % level of significance in both the short- and long-run. A 1 % increase in the interaction between AI and explanatory variables such as overall governance, political governance, and institutional governance resulted in a decrease in growth by 0.028 %, 0.004 %, and 0.015 % in the short-run, and by 0.027 %, 0.003 %, and 0.015 % in the long-run. These results suggest that as AI technology becomes more integrated into these governance aspects, it tends to impede economic growth. This could imply that excessive or misaligned AI interventions in these areas may introduce inefficiencies or disrupt established economic processes, leading to reduced growth prospects in the BRICS economies. Conversely, a 1 % increase in the interaction between AI and economic governance led to an increase in growth by 0.008 % in the short-run and 0.007 % in the long-run. This implies that well-aligned and effective AI applications in economic governance can contribute positively to growth. Such

Table 18
Panel CS-ARDL estimates for economic growth model.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Variables											
Short Run Est.											
ΔLAI	0.001***	−0.033	0.023**	0.004	0.006*	0.039***	0.000	−0.113***	−0.290***	−0.063***	−0.093
	(0.000)	(0.026)	(0.012)	(0.010)	(0.003)	(0.010)	(0.000)	(0.039)	(0.025)	(0.018)	(0.112)
ΔLFDI	−0.076***	0.002*	0.023**	0.004***	−0.018	−0.034	0.057**	−0.005	−0.153	0.040***	0.026**
	(0.024)	(0.001)	(0.011)	(0.001)	(0.028)	(0.062)	(0.025)	(0.011)	(0.105)	(0.014)	(0.012)
ΔLGFCF	0.086	0.001	−0.125***	0.067	0.170**	−0.049	0.413**	0.145***	0.660**	0.021	0.069***
	(0.078)	(0.167)	(0.021)	(0.112)	(0.076)	(0.275)	(0.178)	(0.054)	(0.267)	(0.028)	(0.018)
Δ LHMC	0.154***	1.848***	−0.006	−0.030***	−0.131***	−0.061***	0.047**	0.387***	2.187	0.760	0.503
	(0.051)	(0.600)	(0.752)	(0.010)	(0.035)	(0.017)	(0.018)	(0.072)	(1.317)	(0.530)	(0.825)
ΔOGV	0.042***	0.672**									
	(0.009)	(0.272)									
ΔLAI*OGV		−0.028**									
		(0.011)									
ΔPOLG			0.073*								
			(0.043)								
ΔLAI* POLG			−0.004**								
			(0.002)								
ΔINSTG				0.332*							
				(0.173)							
ΔLAI* INSTG				−0.015*							
				(0.008)							
ΔECOG					−0.134**						
					(0.068)						
ΔLAI* ECOG					0.008***						
					(0.003)						
ΔCRT						0.343**					
						(0.162)					
ΔLAI*CRT						−0.000***					
						(0.000)					
ΔPOLV							−0.002				
							(0.002)				
ΔLAI*POLV							−0.001***				
							(0.000)				
ΔGEF								0.378*			
								(0.210)			
ΔLAI*GEF								−0.020*			
								(0.011)			
ΔRQE									0.966*		
									(0.565)		
ΔLAI*RQE									−0.043*		
									(0.023)		
ΔRLW										1.370**	
										(0.696)	
ΔLAI*RLW										−0.061**	
										(0.031)	
VCAC											3.647***
											(1.259)
ΔLAI*VCAC											−0.183***
											(0.046)
Adjust. Term											
ECT	−1.076***	−1.004***	−1.125***	−1.030***	−1.132***	−1.061***	−1.015***	−1.113***	−1.290***	−1.063***	−1.086***
	(0.024)	(0.003)	(0.021)	(0.010)	(0.035)	(0.017)	(0.007)	(0.039)	(0.025)	(0.018)	(0.028)
Long Run Est.											
LR_LAI	0.001***	−0.032	0.020**	0.003	0.005*	0.038***	0.000	−0.004	−0.006	0.001	−0.091
	(0.000)	(0.026)	(0.011)	(0.009)	(0.003)	(0.010)	(0.000)	(0.007)	(0.039)	(0.001)	(0.106)
LR_LFDI	−0.010	0.002*	0.020**	0.004***	−0.016	−0.031	0.056**	−0.004	−0.119	0.038***	0.024**
	(0.015)	(0.001)	(0.009)	(0.001)	(0.025)	(0.058)	(0.025)	(0.011)	(0.080)	(0.014)	(0.010)
LR_LGFCF	0.081	0.002	0.072	0.067	0.151**	−0.035	0.408**	0.129***	0.516**	0.019	−0.064***
	(0.073)	(0.166)	(0.068)	(0.108)	(0.068)	(0.258)	(0.176)	(0.048)	(0.207)	(0.026)	(0.017)
LR_LHMC	0.141***	1.846***	−0.003	0.449	0.173	0.035	0.046**	0.342***	1.748*	0.725	0.475
	(0.046)	(0.601)	(0.680)	(0.405)	(0.323)	(0.088)	(0.017)	(0.054)	(1.059)	(0.520)	(0.765)
LR_OGV	0.039***	0.668**									
	(0.009)	(0.271)									
LR_LAI*OGV		−0.027**									
		(0.011)									
LR_POLG			0.066*								
			(0.038)								
LR_LAI* POLG			−0.003**								
			(0.002)								
LR_INSTG				0.327*							
				(0.170)							
LR_LAI*INSTG				−0.015*							
				(0.008)							

(*continued on next page*)

Table 18 (*continued*)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
LR_ECOG					−0.120**						
					(0.062)						
LR_LAI* ECOG					0.007***						
					(0.003)						
LR_CRT						0.330**					
						(0.159)					
LR_LAI*CRT						−0.000***					
						(0.000)					
LR_POLV							−0.002				
							(0.002)				
LR_LAI*POLV							−0.001***				
							(0.000)				
LR_GEF								0.315*			
								(0.175)			
LR_LAI*GEF								−0.017*			
								(0.009)			
RQE									0.785*		
									(0.476)		
LR_LAI*RQE									−0.035*		
									(0.019)		
LR_RLW										1.330**	
										(0.674)	
LR_LAI*RLW										−0.060**	
										(0.030)	
LR_VCAC											3.410***
											(1.209)
LR_LAI*VCAC											−0.171***
											(0.045)
Observation	210	210	210	210	210	210	210	210	210	210	210
R-squared	0.75	0.97	0.89	0.72	0.72	0.58	0.95	0.86	0.67	0.82	0.62

Note: Standard errors in parentheses; *, ** and *** represent significance at the 1%, 5%, and 10% levels respectively. The dependent variable is economic growth. Source: Author's Computations.

applications may enhance economic policies, streamline regulatory processes, or optimize resource allocation, thereby fostering an environment conducive to economic expansion in the BRICS economies.

In Column 6, 7, 8, 9, 10 and 11 of Table 18, the results indicate that the interaction between AI and each of the governance indicators such as CRT, POLV, GEF, RQE, RLW and VCAC has a negative and significant impact on growth at a 10 % level of significance in the both the short- and long-run. A 1 % increase in the interaction between AI and explanatory variables such as CRT, POLV, GEF, RQE, RLW and VCAC decreased growth by 0.000 %, 0.001 %, 0.020 %, 0.043 %, 0.061 % and 0.183 % in the short-run and by 0.000 %, 0.001 %, 0.017 %, 0.035 %, 0.060 %, 0.171 % in the long-run. The results reveal that the negative impact of the interaction terms is more pronounced when AI is interacted with GEF, RQE, RLW and VCAC in the short-run when compared with the long-run results. The different levels of corruption in BRICS countries can hinder the effective implementation of AI technologies. Corruption may lead to inefficient allocation of resources, mismanagement of AI projects, and a lack of transparency in decision-making. These factors can undermine the potential benefits of AI for economic growth. The negative impact of the interaction between AI and the rule of law on growth in BRICS countries underscores the importance of establishing clear, supportive legal frameworks for AI adoption.

To promote economic growth, policymakers may need to address legal complexities, streamline regulations, and ensure that the legal environment is conducive to AI innovation and investment in the BRICS countries. Political stability and the absence of violence are essential for building investor and consumer confidence in the BRICS bloc. The result of the interaction between AI and POLV may suggest that countries with political turmoil struggle to attract AI-related investments and may face difficulties in creating a conducive environment for AI-driven growth. These factors could be possible reasons behind the obtained results. Imbalanced regulation, particularly in the context of emerging technologies such as AI, has the potential to stifle innovation and impede economic growth. When regulatory bodies exhibit excessive stringency or fail to promptly adapt to the rapidly evolving AI landscape, it can discourage investment and hinder the adoption of AI technologies intended to drive growth in BRICS countries. This may help explain the negative and significant impact observed in the interaction between AI and regulatory quality on growth, both in the short- and long-run.

The negative and significant impact of the interaction between AI and government effectiveness on growth in both the short- and long-run could be attributed to several factors in the case of BRICS countries, for example: (i) If government effectiveness is low in BRICS countries, bureaucratic inefficiency can prevail. This inefficiency might manifest as lengthy approval processes, red tape, and slow decision-making. These factors could hinder the rapid adoption and integration of AI technologies into public and private sectors, thereby possibly slowing down their potential to contribute to economic growth; (ii) investors and businesses across BRICS countries usually prefer stable and predictable regulatory environments. When government effectiveness is low, there may be greater uncertainty about the future regulatory landscape for AI technologies. This uncertainty can discourage long-term investment in AI projects in BRICS countries. And also, in countries with poor government effectiveness, corruption and mismanagement may be more prevalent. Such an environment can deter foreign and domestic investments in AI-related projects, as investors may perceive a higher risk of funds being misappropriated or mismanaged.

In Column 2, 3, 4 and 5 of Table 19, the results indicate that the interaction between AI and explanatory variables such as overall governance, economic governance and institutional governance has a positive and significant impact on employment at a 10 % level of significance in both the short- and long-run (except political governance). These results imply that the synergy between AI and good governance dimensions/practices (overall, economic governance, and institutional governance, excluding political governance) significantly boosts employment. This positive impact is observable in both the short-term and long-term, emphasizing the role of effective governance dimensions in enhancing AI's employment benefits. In Column 6, 7, 8, 9, 10 and 11 of Table 19, the results indicate that the interaction between AI and each of the governance indicators (that is, CRT, POLV, GEF, RQE,

Table 19
Panel CS-ARDL estimates for the employment model.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Variables											
Short Run Est.											
ΔLAI	0.002***	−0.002*	−0.003*	−0.000	0.004**	−0.001	0.001**	0.016***	0.001**	0.001	0.001***
	(0.001)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
ΔLFDI	0.012**	−0.004	0.003	0.004**	0.005**	0.014**	0.007	0.001	0.002	0.042*	0.004
	(0.002)	(0.003)	(0.015)	(0.002)	(0.002)	(0.005)	(0.005)	(0.005)	(0.008)	(0.039)	(0.005)
ΔLGFCF	−0.023***	0.020	0.002**	−0.014	−0.036*	0.071***	−0.064***	0.063***	0.047**	0.032*	0.040***
	(0.008)	(0.018)	(0.001)	(0.110)	(0.022)	(0.013)	(0.027)	(0.020)	(0.024)	(0.018)	(0.017)
Δ LHMC	0.028**	0.060*	−0.190***	−0.126***	−0.099**	0.098**	−0.128***	−0.101**	0.088**	0.057***	−0.146**
	(0.007)	(0.036)	(0.032)	(0.053)	(0.031)	(0.044)	(0.040)	(0.050)	(0.041)	(0.011)	(0.065)
ΔOGV	0.001***	0.004*									
	(0.000)	(0.002)									
ΔLAI*OGV		0.010**									
		(0.000)									
ΔPOLG			0.008								
			(0.006)								
ΔLAI* POLG			−0.001*								
			(0.000)								
ΔINSTG				−0.269*							
				(0.161)							
ΔLAI* INSTG				0.012**							
				(0.007)							
ΔECOG					−0.046**						
					(0.022)						
ΔLAI* ECOG					0.002**						
					(0.001)						
ΔCRT						0.125***					
						(0.045)					
ΔLAI*CRT						0.007**					
						(0.003)					
ΔPOLV							0.012**				
							(0.006)				
ΔLAI*POLV							0.010***				
							(0.000)				
ΔGEF								−0.255**			
								(0.103)			
ΔLAI*GEF								0.013***			
								(0.005)			
ΔRQE									0.012		
									(0.012)		
ΔLAI*RQE									0.001**		
									(0.000)		
ΔRLW										−0.141*	
										(0.085)	
ΔLAI*RLW										0.006*	
										(0.004)	
VCAC											−0.028***
											(0.007)
ΔLAI*VCAC											0.003
											(0.003)
Adjust. Term											
ECT	−0.996***	−1.012***	−1.007***	−1.015***	−1.055***	−1.059***	−1.098***	−1.040***	−1.074***	−0.998***	−1.342***
	(0.115)	(0.046)	(0.055)	(0.008)	(0.027)	(0.038)	(0.050)	(0.033)	(0.044)	(0.016)	(0.114)
Long Run Est.											
LR_LAI	0.002**	−0.002***	−0.003**	−0.000	0.003**	−0.001	0.001***	0.012***	0.001**	0.001	0.001***
	(0.001)	(0.046)	(0.001)	(0.000)	(0.001)	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.000)
LR_LFDI	0.001	−0.004	0.003	0.004	0.013**	0.015**	0.006	0.001	0.002	0.040**	0.002
	(0.002)	(0.003)	(0.015)	(0.002)	(0.001)	(0.004)	(0.004)	(0.005)	(0.007)	(0.036)	(0.003)
LR_LGFCF	−0.029**	0.020	0.002**	−0.013	−0.033*	0.065***	−0.056**	0.060**	−0.041**	0.033*	0.028***
	(0.015)	(0.019)	(0.001)	(0.108)	(0.020)	(0.020)	(0.024)	(0.019)	(0.020)	(0.020)	(0.012)
LR_LHMC	0.036**	0.058***	−0.191***	−0.124**	−0.094***	0.092**	−0.121***	0.097**	−0.085**	0.057***	−0.109**
	(0.005)	(0.006)	(0.034)	(0.053)	(0.029)	(0.023)	(0.039)	(0.047)	(0.041)	(0.011)	(0.053)
LR_OGV	0.001**	0.004*									
	(0.000)	(0.002)									
LR_LAI*OGV		0.011*									
		(0.000)									
LR_POLG			0.008								
			(0.006)								
LR_LAI* POLG			−0.001**								
			(0.000)								
LR_INSTG				−0.267*							
				(0.160)							
LR_LAI*INSTG				0.012*							
				(0.007)							

(*continued on next page*)

Table 19 (*continued*)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
LR_ECOG					−0.044**						
					(0.022)						
LR_LAI* ECOG					0.002**						
					(0.001)						
LR_CRT						−0.117***					
						(0.043)					
LR_LAI*CRT						0.006***					
						(0.003)					
LR_POLV							0.011*				
							(0.006)				
LR_LAI*POLV							0.009***				
							(0.000)				
LR_GEF								−0.242**			
								(0.004)			
LR_LAI*GEF								0.013***			
								(0.099)			
RQE									0.011***		
									(0.001)		
LR_LAI*RQE									0.001**		
									(0.000)		
LR_RLW										−0.139*	
										(0.082)	
LR_LAI*RLW										0.006*	
										(0.004)	
LR_VCAC											−0.022***
											(0.006)
LR_LAI*VCAC											0.002
											(0.002)
Observation	210	210	210	210	210	210	210	210	210	210	210
R-squared	0.99	0.85	0.91	0.89	0.80	0.85	0.90	0.73	0.86	0.64	0.86

Note: Standard errors in parentheses; *, ** and *** represent significance at the 1%, 5%, and 10% levels respectively. The dependent variable is employment. Source: Author's Computations.

RLW and VCAC) has a positive and significant impact on employment at a 10 % level of significance in the both the short- and long-run. These results imply that the interaction of AI with governance indicators (like CRT, POLV, GEF, RQE, RLW, VCAC) has a consistently positive and significant effect on employment, both in the short and long run, indicating that effective governance combined with AI can significantly bolster employment prospects.

5. Conclusion and policy recommendations

5.1. Conclusion

The BRICS countries, namely Brazil, Russia, India, China, and South Africa, are committed to achieving and sustaining United Nations Sustainable Development Goals (SDGs), specifically Goal 8, which focuses on promoting sustainable economic growth and productive employment and decent work for all. To attain this goal, it is crucial to leverage the explanatory variables identified in this study. This study examines the impact of artificial intelligence (AI) on employment and economic growth contingent on governance quality/indicators for BRICS countries between 2012 and 2022. We applied the novel Cross-Sectional Augmented Autoregressive Distributed Lag (CS-ARDL) estimation and other novel econometric techniques. This research is unique compared to previous literature because it examines the causality relationship between AI, economic growth and different dimensions of governance quality. It further examines the impact of the interaction between AI and other explanatory variables such as overall governance, political governance, institutional governance, economic governance, and each of the governance indicators on employment and economic growth. This is because globally, regionally, and within economic blocs, the investment in and the use of AI services/infrastructures cannot operate outside the governance structure of a particular economy. Consequently, formulating integrated policies that are informed by the findings of this study has become critical in the BRICS context.

This study contributes to the literature by yielding the following results. Firstly, it establishes a long-run equilibrium relationship among the variables for both models—the employment and growth models. Secondly, the causality results indicate unidirectional causality running from (i) growth to AI and (ii) AI to employment, among other findings. Thirdly, regarding the long-run coefficient estimates, the outcomes from both FMOLS and DOLS methodologies indicate that: (i) AI and overall governance quality exhibit a significant and positive impact on growth; (ii) the interaction between AI and overall governance quality (OGV) demonstrates a significant and positive impact on growth; (iii) AI and overall governance quality show a significant and negative impact on employment; (iv) the interaction between AI and overall governance quality (OGV) displays a significant and positive impact on employment, among other findings.

Fourthly, the CS-ARDL results for the growth model reveal that: (i) AI and overall governance quality have a significant and positive impact on growth both in the short- and long-run; (ii) the results indicate that the interaction between AI and explanatory variables such as overall governance, political governance, and institutional governance has a negative and significant impact on growth both in the short- and long-run, among other findings. Fifthly, the CS-ARDL results for the employment model reveal that: (i) AI and overall governance quality have a significant and positive impact on employment both in the short- and long-run; (ii) the results indicate that the interaction between AI and explanatory variables such as overall governance, institutional governance and economic governance has a positive and significant impact on employment both in the short- and long-run, among other findings.

Our research is crucial for the strategic development of AI and governance policies designed to promote employment and economic performance in BRICS countries. This is because AI interaction with governance indicators must be taken seriously to ensure and productive employment and decent work for all. Furthermore, to maximise the mediating effects of governance quality/dimensions/indicators on the impact of AI on employment and economic growth in BRICS economies, policymakers should prioritize context-specific implementation, ethical considerations, robust governance, integration with existing systems,

capacity building, risk assessment, and public engagement.

5.2. Policy recommendations

The significant and positive impact of AI and overall governance quality on growth in both the short- and long-run has crucial policy implications. For instance, the BRICS governments and policymakers should recognize the potential of AI and good governance practices as drivers of employment and economic growth. Encouraging investments in AI technology and cultivating a robust governance environment, marked by political stability, adherence to the rule of law, and supportive economic policies, among other factors promoting innovation, will prove advantageous for both employment and economic development in the BRICS nations. Additionally, policymakers should consider both short-term and long-term effects of AI and governance quality on employment and growth when designing strategies to promote employment and economic growth.

BRICS governments and policymakers should aim for a balanced integration of AI into their distinct governance structures to foster employment and growth. This is crucial, considering that the interaction between AI and governance dimensions/indicators yields mixed results (both positive and negative), impacting employment and growth in both the short- and long-run. Therefore, it's crucial to carefully formulate, assess, manage, and review the extent of AI policy integration in these areas to monitor their potential threat to employment and economic growth. The positive impact of the interaction between AI and economic governance on employment and growth suggests that BRICS countries should implement policies that encourages active investment in AI technologies that can optimize employment and economic policies, enhance regulatory efficiency, and improve resource allocation, all with the aim of creating more job opportunities and promoting growth. Policies aimed at promoting innovation and the responsible adoption of AI, in conjunction with economic governance, should be encouraged.

We recommend that BRICS countries enhance the quality of governance practices, including addressing issues related to the six governance indicators. They should also strive to create governance environments capable of effectively incorporating and adapting to AI technologies without impeding jobs and growth. This can be achieved in part by building the capacity of government officials and institutions to comprehend, implement, and regulate AI technologies. Such efforts are crucial, as training programs and knowledge-sharing initiatives among BRICS member states can facilitate the effective adoption of AI while mitigating any potential negative consequences that may arise from its interaction with governance indicators/dimensions on employment and economic growth. Policymakers should ensure that policies related to AI align with broader governance dimensions/indicators. This should involve cross-governance dimensions/indicators coordination to prevent joint conflicting AI applications and governance indicators that could harm employment and growth. Policies that encourage regular monitoring and evaluation of AI integration into governance structures for employment and growth objectives are essential and should be enacted and implemented. This will enable adjustments in AI strategies, both in the short- and long-term, to ensure their continued alignment with the employment and economic growth policy objectives of BRICS countries. BRICS countries should involve stakeholders in AI investment decision-making processes, ensuring that the public understands the rationale behind AI integration into governance dimensions/indicators and its potential impacts on employment and economic growth.

5.3. Limitations of the study

Future research should examine whether the established conclusions in this study hold up to empirical inspection within country-specific or regional settings to further enhance our current understanding of the research topic. This will be particularly pertinent for deriving more country-specific or regional-level policy implications.

CRediT authorship contribution statement

Charles Shaaba Saba: Conceptualization, Data curation, Formal analysis, Investigation, Resources, Software, Validation, Writing – original draft, Writing – review & editing. **Nicholas Ngepah:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability.

All data generated or analysed during this study are not included in this submission but can be made available upon reasonable request. Additionally, the data are publicly available.

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Author's contributions

CSS and NN conceptualised the study idea, drafted the paper, collected data, analysed data, wrote the introduction section, organised the literature review, drafted the methodology section, interpreted the results and provided the discussions, concluded the study with policy implications, and organised the reference list. CSS and NN equally proofread, edited, and finalised the article.

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